


长春理工大学

Changchun University of Science and Technology

# 课程期末论文

# Gaussian Beam Propagation and Focusing: Principles and Applications

FINAL COURSE DISSERTATION

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# Abstract

This dissertation examines Gaussian laser-beam propagation and focusing by combining analytical optics, Python calculations, and analysis of existing laser-beam photographs. In the first experiment, I applied the paraxial $TEM_{00}$ model to three parameter studies: changing wavelength at a fixed waist, changing the initial waist at a fixed wavelength, and changing a thin lens and its incident beam. The calculations report beam radius, Rayleigh range, divergence, focused waist location, and focused spot size. At 1064 nm and an initial waist of 0.50 mm, the ideal model gives a Rayleigh range of 0.738 m and a divergence half-angle of 0.677 mrad. The simpler focal-spot approximation overestimates the exact waist by 11.80% when $f/zR = 0.50$ in the waist-at-lens configuration.

In the second experiment, I analyzed 14 $TEM_{00}$ beam images provided with the DataLab laser-beam-profiling tutorial. The recorded JPEG images were not collected by me. For each image, the processing script estimated a beam center, extracted horizontal and vertical intensity profiles, and fitted a Gaussian with an adjustable background. The fitted $1/e^2$ intensity radii ranged from 163.83 to 297.96 pixels horizontally and from 164.97 to 304.00 pixels vertically; the fitted profile $R^2$ values were 0.9927–0.9959. The image sequence shows a broadening trend and profiles that are well described by the chosen Gaussian model. However, the available files do not establish the physical pixel scale, wavelength, or calibrated axial positions needed to infer an experimental Rayleigh range or beam-quality factor.

The combined work demonstrates two different uses of Gaussian-beam theory: predicting ideal optical behavior from known parameters and characterizing measured intensity patterns from images. Their distinction is essential when moving from optical formulas to experimental interpretation.



Keywords: Gaussian beam; $TEM_{00}$; beam propagation; Rayleigh range; divergence; ABCD matrix; thin-lens focusing; intensity-profile fitting; image analysis

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# List of Symbols and Abbreviations

| | |
|---|---|
| $TEM_{00}$ | Fundamental transverse electromagnetic Gaussian mode |
| ABCD | Paraxial ray-transfer matrix coefficients |
| $w_0$ | Minimum $1/e^2$ intensity radius (beam waist) |
| $w(z)$ | Beam radius at axial position z |
| $z_R$ | Rayleigh range |
| $\lambda$ | Optical wavelength in the propagation medium |
| $\theta$ | Far-field divergence half-angle |
| $R(z)$ | Wavefront radius of curvature |
| q | Complex Gaussian beam parameter |
| f | Thin-lens focal length |
| $M^2$ | Beam propagation ratio / beam quality factor |
| FSO | Free-space optical communication |
| FWHM | Full width at half maximum |
| $R^2$ | Coefficient of determination for an intensity profile fit |
| px | Image pixel (uncalibrated spatial unit) |

# Chapter 1: Introduction

## 1.1 Research background

Laser light is highly directional, but a beam with a finite cross-section cannot remain perfectly parallel as it travels. Diffraction causes its transverse profile to evolve, even in uniform free space. For a beam close to the fundamental transverse mode, the Gaussian description captures this behavior through three closely related quantities: the minimum beam radius, the growth of that radius with distance, and the curvature of the optical wavefront. These ideas provide the starting point for analyzing laser propagation and focusing [1, 2].

Unlike a purely geometric ray model, Gaussian-beam optics simultaneously incorporates a beam radius, a wavefront radius of curvature, and a longitudinal phase shift. The resulting framework provides a practical link between the wave equation and optical design. Siegman's treatment of laser beams and resonators established a comprehensive classical reference for understanding these properties and their significance for laser systems [1]. The same principles support the design of output optics, diagnostic instruments, beam expanders, communication terminals, and focused-laser processing arrangements.

The engineering task is not merely to send light from a laser to a target. Often the beam must reach the target with a specified diameter, intensity profile, depth of focus, or angular spread. A narrow focal spot can increase irradiance on a surface, but it also produces a shorter region over which the beam remains relatively narrow. Conversely, a beam designed for low divergence requires a larger characteristic waist and may be less appropriate for tight focusing. This interplay makes Gaussian-beam propagation and focusing a suitable analytical study rather than a list of unrelated laser applications. Figure 1.1 illustrates the corresponding beam envelope for one stated set of parameters; the curve is calculated from the ideal model and is not measured data.

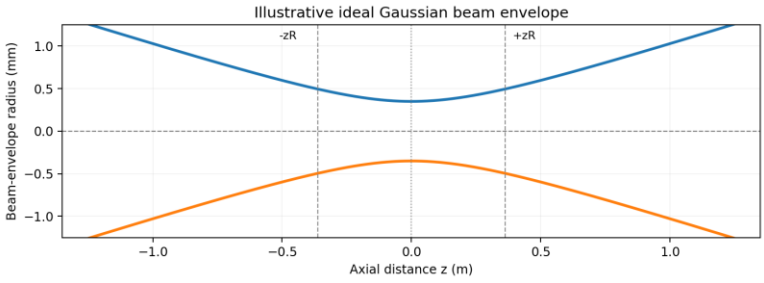



Figure 1.1 Calculated Gaussian beam envelope ($\lambda$ = 1064 nm, $w_0$ = 0.35 mm); ideal paraxial model.

## 1.2 Problem statement and motivation

The familiar ray diagram of a converging lens suggests that parallel light focuses at the focal plane. A Gaussian beam is more complicated. The light arriving at the lens may already have a curved wavefront, so both the new waist position and its radius depend on

the incident beam as well as the lens. The complex beam parameter provides a convenient way to keep track of these effects through optical components [2].

A second difficulty is interpreting a beam's size correctly. The Gaussian radius is conventionally defined at the transverse position where intensity has fallen to $1/e^2$ of its on-axis value, rather than at a physical boundary. The radius depends on axial position and is linked to wavelength through the Rayleigh range. A parameter change that improves one property may weaken another: reducing the input waist, for example, strengthens far-field divergence at fixed wavelength. Any numerical comparison must therefore state its fixed variables and optical assumptions explicitly.

The question considered here is how the main Gaussian-beam parameters work together in a simple optical system. The analysis proceeds from the fundamental field distribution to free-space propagation, then to lens transformations and calculated examples. This makes it possible to compare focusing choices using the same definitions and assumptions throughout.

## 1.3 Research question and objectives

The main research question is: How do wavelength, input beam waist, and lens focal length control the propagation and focusing of an ideal Gaussian beam, and what can Gaussian fitting tell us about the transverse intensity patterns in an existing collection of beam photographs?

The study pursues the following specific objectives:

- Establish the physical interpretation and mathematical definitions of intensity profile, beam waist, Rayleigh range, and divergence.
- Derive and interpret the free-space propagation relations for an ideal, monochromatic TEM00 beam.
- Apply the complex beam parameter and ABCD matrix law to free-space intervals and thin lenses.
- Compare predicted focal spot radii and waist locations across reproducible parameter choices.
- Explain the engineering consequences of the spot-size/divergence trade-off and identify the limitations of the ideal model.
- Estimate horizontal and vertical Gaussian intensity radii from a published image sequence, assess fit quality, and state which physical quantities remain unidentifiable without calibration.

## 1.4 Scope and assumptions

The analytical calculation assumes scalar, paraxial, monochromatic propagation in homogeneous free space and an ideal circular $TEM_{00}$ mode ($M^2 = 1$). The thin-lens calculations neglect clipping, aberration, scattering, thermal effects, nonlinearities, and

misalignment. The separate image-based study uses a third-party set of beam-profile JPEGs. Unlike the simulated results, those profiles are derived from recorded intensity distributions; their fitted widths are reported only in image pixels because transverse spatial calibration and independently verified axial distances are not supplied with the processed outputs.

Real laser outputs may be asymmetric, partially clipped, saturated in camera images, or composed of higher-order modes. For this reason, the two studies are presented separately: the numerical plots are ideal model predictions, and the DataLab-image fits describe observable pixel profiles. Neither study establishes a calibrated propagation factor $M^2$ for a specific laser. No physical experiment was performed by the author.

## 1.5 Research methodology

The dissertation combines two complementary methods. First, the standard paraxial Gaussian equations and the complex-q/ABCD rule are developed, then evaluated in a reproducible Python numerical study. The wavelength sweep, initial-waist sweep, and thin-lens sweep form Experiment 1; their inputs are selected theoretical cases, not measured samples. Experiment 2 starts from 14 image files published in the DataLab laser-beam-profiling tutorial [12]. Each image is converted to a grayscale intensity array, its background and approximate center are estimated, horizontal and vertical profiles are extracted, and four-parameter Gaussian curves are fitted by nonlinear least squares. Results are compared through fitted radii, FWHM, goodness of fit, and changes across the ordered image sequence.

For the first experiment, all wavelengths, waists, focal lengths and propagation distances are converted to SI units before calculation. Analytical checkpoints include $w(0) = w_0 \; and \; w(zR) = \sqrt{2}w_0$. For the second, the program records its fitted radii in pixels and checks the quality of one-dimensional profile fits using $R^2$. The two coordinate systems are not interchanged: the image labels are used to order the files, not treated as calibrated propagation distances. The original photographs were not supplied with independently verified camera-pixel pitch or laser wavelength in this study.

## 1.6 Dissertation organization

Chapter 2 reviews the relevant optical theory and beam measurement conventions. Chapter 3 develops the fundamental Gaussian field, followed by free-space propagation in Chapter 4 and ABCD lens transformations in Chapter 5. Chapter 6 presents Experiment 1 (three numerical studies) and Experiment 2 (analysis of 14 published images), with separate methods and results before their joint discussion. Chapter 7 connects both results to optical applications and measurement limits. Chapter 8 states the conclusions and work needed for a calibrated experimental test.

# Chapter 2: Literature Review

## 2.1 Development of the Gaussian-beam framework

Gaussian-beam theory is part of the broader development of wave optics and resonator analysis. In the scalar paraxial approximation, a slowly varying complex envelope is superimposed on a carrier wave. The fundamental solution has a Gaussian transverse amplitude, while its longitudinal evolution can be summarized through beam radius, phase-front curvature, and the Gouy phase. This framework is particularly useful because it retains the consequences of diffraction without requiring a full electromagnetic numerical solution for each elementary optical layout.

The development of Gaussian optics links diffraction theory with the design of practical laser systems. Siegman gives a comprehensive treatment of laser modes, resonators, and beam propagation [1], while Kogelnik and Li explain the behavior of laser beams in resonators and first-order optical systems [2]. Related work on propagation in lens-like media and resonators containing lenses establishes the usefulness of the complex beam parameter [3, 4]. Together, these contributions provide a clear basis for treating beam size and wavefront curvature as coupled optical quantities.

The optical literature treats the fundamental Gaussian mode as the baseline from which more elaborate descriptions can be built. Higher-order Hermite–Gaussian modes, astigmatic beams, apertured beams, and non-paraxial fields may require additional parameters or different methods. As a consequence, statements about an ideal Gaussian spot should not automatically be generalized to every laser output. This dissertation uses the ideal solution as an explicitly defined reference case before discussing departures from it.

## 2.2 Complex beam parameter and ABCD transformations

A major strength of Gaussian optics is the ability to express beam propagation through simple optical components using a single complex parameter q. It incorporates wavefront curvature and spot size, and its transformation through any first-order optical arrangement can be written as $qout = (Aqin + B)/(Cqin + D)$, where A, B, C, and D are the elements of the ray-transfer matrix. The utility of this representation lies in its composability: multiple propagation distances and optical elements can be represented by matrix multiplication.

Kogelnik's 1965 analysis of Gaussian beams in lens-like media showed how a complex beam parameter simplifies propagation in media whose refractive index or gain profile varies quadratically with transverse position [3]. In the same period, Kogelnik's work on resonators containing internal lenses developed related transformation rules for lenses, telescopes, and sequences of elements [4]. These studies help explain why the ABCD

method is not an arbitrary mathematical shortcut: it is a compact representation of paraxial beam evolution under a defined set of physical approximations.

For this dissertation, the matrix approach will first be applied to two canonical components: free-space travel over distance L, represented by the matrix $[1\ L;\ 0\ 1]$, and an ideal thin lens with focal length f, represented by $[1\ 0;\ -1/f\ 1]$. Matrix products will be formed in optical-traversal order. The resulting q parameter will be used to recover the local beam radius and wavefront curvature. Later numerical examples will check that the free-space limit agrees with the usual closed-form propagation law.

## 2.3 Research on Gaussian-beam focusing

Gaussian-beam focusing differs from ordinary ray imaging because a lens changes the phase-front curvature without changing the beam radius instantaneously at the lens plane. Only after further propagation does the field reach its new minimum radius. The incident beam parameter and the focal length therefore determine both the waist location and its size. This distinction is central to Gaussian lens calculations [2, 5].

Self studied the focusing of spherical Gaussian beams and described procedures for tracing them through trains of lenses or mirrors [5]. This work is a helpful bridge between conventional lens design and Gaussian-beam design because it treats the position and properties of a beam waist as optical variables. The distinction matters when an incident beam waist is not located at the lens or when the incident beam has appreciable curvature.

The familiar estimate for a tightly focused, nearly collimated ideal beam, $wfocus \approx \lambda f/(\pi wL)$, is useful but conditional. Here wL denotes the beam radius at the lens. The approximation is best viewed as a design rule under weak incident convergence or divergence, negligible aperture clipping, and negligible aberration. A general q-parameter transformation is preferable when those geometrical conditions are not satisfied. The dissertation will include both approaches and compare their use within appropriate limits.

## 2.4 Connections to broader wave and photonic optics

The Gaussian solution can also be understood in the context of scalar diffraction and Fourier optics. Because spatial confinement and angular spread are related, a small waist is associated with a relatively broad angular spectrum. Goodman's treatment of diffraction and optical propagation provides the broader Fourier-optics perspective for interpreting such spatial-domain trade-offs [6]. This perspective reinforces the need to account for diffraction when a simple ray diagram suggests that a beam could be confined arbitrarily tightly.

Saleh and Teich's photonics reference places wave optics, ray optics, and electromagnetic optics within a common technological framework [7]. For laser

engineering, that layered viewpoint is valuable: geometric models are often sufficient to select approximate optical positions, but wave models are required to predict diffraction-limited spot sizes. The present study therefore uses matrices from first-order optics while retaining Gaussian wave-optical quantities in the beam parameter.

Combining Gaussian wave optics with ABCD matrices is useful because the same calculation can describe several common arrangements: propagation between optical elements, focusing by a single lens, and collimation by a lens pair. The method is compact, provided that its paraxial and ideal-element assumptions remain appropriate.

## 2.5 Beam-quality measurement and model limitations

The ideal TEM00 field is not a complete model of real beams. An actual laser may have phase distortions, multimode structure, astigmatism, or an intensity profile that does not follow a perfect circular Gaussian. In those cases, a fitted $1/e^2$ radius may not characterize the field sufficiently, and different measurement methods can lead to different reported beam widths. Quantitative comparisons must therefore state which width definition and beam-quality convention are used.

ISO 11146-1:2021 provides standardized methods to measure beam widths, divergence angles, and beam propagation ratios for stigmatic and simple astigmatic beams [8]. Its existence demonstrates that beam radius and divergence are measurement quantities requiring operational definitions rather than merely labels on a diagram. The broader beam-quality factor $M^2$ will be introduced as an engineering correction and diagnostic concept, without implying that every non-ideal beam can be modeled by simply replacing $\lambda$ with $M^2\lambda$ in an arbitrary optical arrangement.

Finite apertures, aberrations, and misalignment also limit achievable focus. Aperture truncation may cause diffraction ripples and reduce transmitted power; aberrations alter the optical phase and can enlarge the minimum usable spot. These effects become increasingly important when a design targets a very small focal radius or when optical components have constrained clear apertures. The present study will discuss such limitations qualitatively and distinguish them from numerical predictions made under the ideal Gaussian model.

## 2.6 Research positioning and intended contribution

The main interest of the present analysis lies in comparing the consequences of familiar Gaussian-beam equations under consistent conditions. Examining wavelength, input waist, and focal length together reveals which changes improve spot size and which increase divergence or reduce working tolerance. It also makes the limitations of simple focusing estimates easier to see.

Particular attention is given to the quadratic dependence of Rayleigh range on waist radius, the inverse dependence of divergence on that radius, and the change in focused spot size with lens geometry. Calculated curves are discussed in physical units and related to representative beam-delivery and focusing problems.

Image-based Gaussian fitting is closely related to optical beam profiling but is not equivalent to full beam-quality metrology. A camera captures a spatially sampled intensity pattern, and fitting a Gaussian profile can estimate its center and $1/e^2$ radius. Standardized propagation-factor measurements require calibrated size measurements at known axial positions and a defined width convention; for example, ISO 11146-1:2021 specifies beam-width, divergence, and propagation-ratio procedures [8]. DataLab provides a public image series and documents profile extraction, background control, fitting, and FWHM calculation [12]. The image workflow in this dissertation is a separate analysis of those photographs using its own stated fitting procedure, not a repetition of DataLab's complete software workflow.

## 2.7 Literature review summary

The literature supports three conclusions that guide the subsequent chapters. First, the ideal Gaussian field is a well-founded paraxial wave solution that captures both transverse confinement and diffraction-driven spreading. Second, the complex q parameter and ABCD matrices enable concise transformations through practical optical systems. Third, beam measurements and achievable focal performance depart from the ideal model whenever finite apertures, aberrations, or non-Gaussian structure become significant. These conclusions define both the strength and the boundaries of the theoretical and numerical investigation developed next.

# Chapter 3: Fundamentals of Gaussian Beams

## 3.1 Wave-optical origin of the Gaussian description

The transverse structure of a finite optical beam changes during propagation because it contains more than one transverse spatial frequency. Those components accumulate phase at different rates, leading to diffraction and beam spreading. In the paraxial limit, the scalar wave equation has a fundamental Gaussian solution that describes this effect without requiring a separate full-field calculation for every propagation distance [1, 2].

For monochromatic light propagating through a homogeneous, isotropic, lossless medium with wavenumber $k = 2\pi/\lambda$, consider the scalar Helmholtz equation. Here λ is the wavelength in the medium, not necessarily the vacuum wavelength. Refractive-index variations and polarization coupling are neglected at this stage. The scalar treatment is adequate for a fundamental beam whose propagation angles are small and whose characteristic radius is much larger than the optical wavelength [1, 6].

$$\nabla^2 E + k^2 E = 0, \qquad k = 2\pi / \lambda \tag{3-1}$$

A rapidly oscillating longitudinal carrier can be separated from a slowly varying envelope: $E(x,y,z) = u(x,y,z)\, exp(-i\, k\, z)$. Inserting this into (3-1), and neglecting the second derivative $\partial^2 u/\partial z^2$ relative to 2k $\partial u/\partial z$, gives the paraxial scalar wave equation. The approximation implies $|\partial^2 u/\partial z^2| \ll |2k\, \partial u/\partial z|$ and is not guaranteed close to a focus approaching wavelength-scale dimensions.

$$\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 - 2\, i\, k\, (\partial u/\partial z) \approx 0 \tag{3-2}$$

The rotationally symmetric solution of Equation (3-2) combines a Gaussian amplitude with a quadratic phase term. The amplitude controls the transverse intensity profile, while the phase term represents wavefront curvature. A further longitudinal correction, the Gouy phase, becomes especially important near the waist. All three features follow from the same wave-optical description [1, 2].

## 3.2 Fundamental TEM00 field and intensity

Let $r = \sqrt{(x^2 + y^2)}$ denote the radial distance from the optical axis, and let the plane $z = 0$ coincide with the beam waist. Under a consistent forward-propagating phase convention, the complex electric-field envelope may be written using three distance-dependent functions: beam radius w(z), wavefront curvature radius R(z), and Gouy phase ψ(z).

$$u(r,z) = u_0\, [w_0 / w(z)]\, exp[-r^2 / w^2(z)] \times exp[-i\, k\, r^2 / (2\, R(z)) + i\, \psi(z)] \tag{3-3}$$

The phase convention in (3-3) corresponds to a carrier $exp(i\omega t - i\,k\,z)$. With other accepted sign conventions both the carrier and the curvature-phase sign change consistently; measurable intensity and the beam-transform formulas are unchanged. At the waist, $w(0) = w_0$, R(0) approaches infinity, and $\psi(0) = 0$. The fundamental distribution is often called TEM00 or Gaussian fundamental mode, although the exact modal designation depends on the optical system.

Intensity is proportional to $|u|^2$, so the transverse Gaussian profile follows immediately. For total power P crossing a plane normal to the propagation direction, radial integration provides the normalized power expression in (3-4). This formula assumes a circularly symmetric ideal beam and no absorption or aperture clipping [1, 2].

$$I(r,z) = [2P / (\pi\, w^2(z))]\, exp[-2\, r^2 / w^2(z)] \tag{3-4}$$

Equation (3-4) supplies an operational definition of beam radius: at $r = w(z)$, intensity is $exp(-2) \approx 0.1353$ of its on-axis value. The beam edge is not an abrupt boundary; the Gaussian tail extends mathematically to any transverse distance. This distinction matters when specifying optical apertures, detectors, and beam-width measurement methods.

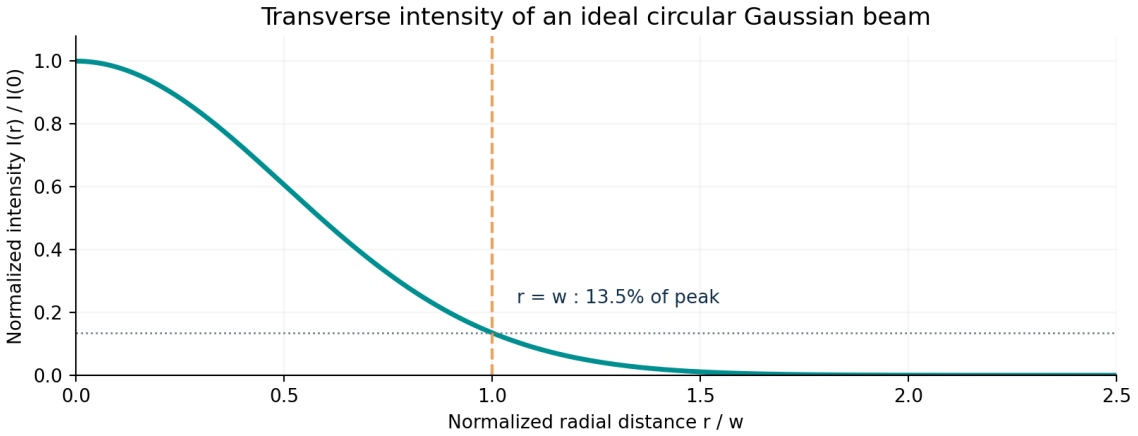



Figure 3.1 Radial intensity of an ideal circular Gaussian beam, evaluated from Equation (3-4) and normalized to the axial intensity.

The encircled power within a circular aperture of radius a is found by integrating $2\pi r I(r,z)$ from 0 to a. The result is useful for distinguishing a waist radius from an aperture radius: a circular aperture equal to one local 1/e² radius passes about 86.5% of the ideal Gaussian power, provided the aperture is centered and located perpendicular to the optical axis.

$$P(r \le a) / P = 1 - exp[-2\, a^2 / w^2(z)] \tag{3-5}$$

For instance, setting $a = w(z)$ in (3-5) yields $1 - e\text{^}(-2) \approx 0.8647$; setting $a = 1.5w(z)$ yields $1 - e\text{^}(-4.5) \approx 0.9889$. The same physical radius can therefore represent different power-capture fractions at different axial locations if the beam is

expanding. This result follows from ideal Gaussian theory and does not model the diffracted field after an aperture.

## 3.3 Beam waist, Rayleigh range, and spot-radius evolution

The beam waist $w_0$ is the smallest $1/e^2$ intensity radius along the optical axis. If $z = 0$ is placed at that waist, free-space diffraction determines the beam radius at every other axial position. The characteristic distance $z_R$ is called the Rayleigh range; it sets the scale over which the beam stays relatively narrow [1, 2].

$$w(z) = w_0 \sqrt{[1 + (z / zR)^2]} \tag{3-6}$$

$$zR = \pi w_0^2 / \lambda \tag{3-7}$$

At $z = \pm zR$ the radius becomes $\sqrt{2}w_0$. Because the area associated with a circular beam radius is proportional to $w^2$, the characteristic area doubles at those planes. At fixed total beam power, the on-axis intensity decreases to half its waist value. The distance between the two Rayleigh planes is $2z_R$ and is also called the confocal parameter. These definitions provide a convenient boundary between the near-waist region and the beginning of appreciable diffraction spreading.

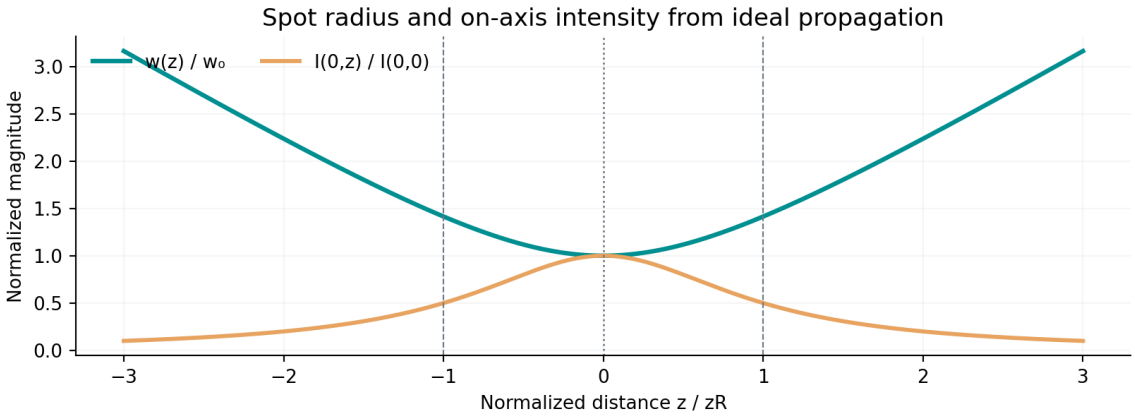



Figure 3.2 Ideal spot-radius growth and on-axis intensity decrease as functions of distance normalized to Rayleigh range (Equations (3-4) and (3-6)).

The waist is not a pinhole or a physical component. It is a geometrical and wave-optical property of the field. A waist may occur in free space, at a focus formed by a lens, or at a reference plane inferred by extending a measured beam backwards. For a beam inside a uniform refractive medium, $\lambda$ in (3-7) is the wavelength within that medium. Using the vacuum wavelength while retaining a medium-dependent $w_0$ without accounting for refractive index would make the prediction inconsistent.

## 3.4 Wavefront curvature and Gouy phase

The fundamental field carries phase information in addition to intensity. The wavefront curvature R(z) is infinite at the waist, decreases to a minimum positive value of $2zR$ at $z = +zR$, and grows approximately like z in the far field. The sign of R distinguishes converging from diverging portions of the beam under the specified propagation convention. A real, finite beam thus does not propagate like a plane wave everywhere, even if its radius varies only slowly near the waist.

$$R(z) = z\,[1 + (zR / z)^2], \quad z \neq 0; \quad R(0) = \infty \tag{3-8}$$

The Gouy phase is a longitudinal phase shift associated with transverse confinement. For an ideal circular TEM00 beam the waist-centered expression is ψ(z) = arctan(z/zR). It changes by π from far before the waist to far after it, and changes rapidly near the focus. This phase correction can influence interferometric and resonant optical systems, even though it is not directly visible in a plot of beam intensity.

$$\psi(z) = arctan(z / zR) \tag{3-9}$$

Rayleigh range, curvature, and Gouy phase are linked consequences of the same Gaussian solution rather than unrelated empirical parameters. The curvature tends to infinity exactly at the waist where the phase is locally planar; the transverse radius is minimized at that same position. These facts also explain why the complex beam parameter introduced in Chapter 5 can package the size and phase-front shape into one quantity.

## 3.5 Gaussian modes beyond TEM00 and the assumptions of this study

A Gaussian beam need not be confined to the fundamental $TEM_{00}$ mode. Higher-order Hermite–Gaussian modes can contain several bright regions separated by dark nodal lines, with transverse phases unlike those of a single circular Gaussian peak [1, 7]. To keep the propagation and focusing comparisons unambiguous, the calculations below use the circular fundamental mode. Higher-order and distorted beams require a more detailed description of their transverse structure.

This restriction also means that ideal-mode formulas should not be used to conceal beam distortions, finite-aperture diffraction, asymmetry, or nonparaxial effects. They are a baseline model against which actual optical systems can be evaluated, rather than a guarantee of diffraction-limited performance in every laser. The distinction will be taken up explicitly in Chapter 7, with reference to standardized beam-width and beam-quality measurement [8].

## 3.6 Chapter summary

Starting with the paraxial wave equation, this chapter defined a circularly symmetric Gaussian field and its intensity distribution, established the $1/e^2$ beam-radius convention, and related the waist radius to diffraction-driven Rayleigh range. Wavefront curvature and Gouy phase complete the fundamental description. The next chapter uses these relations to analyze free-space propagation and to quantify how wavelength and waist size control divergence.

# Chapter 4: Gaussian Beam Propagation Analysis

## 4.1 Normalized beam-envelope evolution

Propagation of an ideal fundamental Gaussian beam through a uniform, lossless medium is governed by the relations derived in Chapter 3. The central prediction is the smooth increase of the $1/e^2$ radius away from its waist, regardless of propagation direction. Expressing distance in units of zR gives a universal curve independent of the specific values of wavelength or initial waist: $w(z)/w_0 = \sqrt{[1 + (z/zR)^2]}$. Therefore, two beams with very different physical scales exhibit the same normalized envelope if they are both ideal paraxial Gaussians [1].

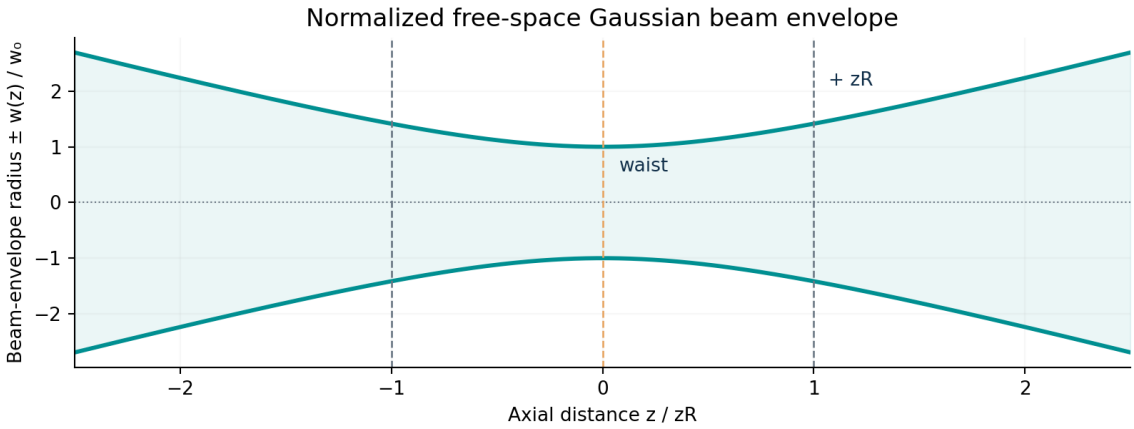



Figure 4.1 Normalized free-space Gaussian envelope. The solid curves are $\pm w(z)/w_0$; dashed vertical lines mark $z = \pm zR$.

Near the waist, $|z| \ll zR$, the binomial approximation predicts that the radius changes only quadratically with axial distance. In contrast, when $|z| \gg zR$, the radius increases almost linearly and its slope approaches the far-field divergence half-angle. This separation explains why a beam may appear essentially collimated across a short optical bench but diverge perceptibly after many Rayleigh lengths.

$$Near\ waist: w(z) \approx w_0\,[1 + z^2 / (2\,zR^2)] \quad for\ |z| \ll zR \tag{4-1}$$

$$Far\ field: w(z) \approx (w_0 / zR)\,|z| = \theta\,|z| \quad for\ |z| \gg zR \tag{4-2}$$

## 4.2 Far-field divergence and the waist-size trade-off

The asymptotic spread of the ideal Gaussian envelope defines the half-angle divergence θ, with the full angle approximately 2θ when paraxial angles are small. Substitution of the Rayleigh range (3-7) into (4-2) yields the familiar inverse relation between divergence and minimum radius. The distinction between full- and half-angle matters: quoted manufacturer divergences and course examples are not always expressed in the same convention.

$$\theta = w_0 / zR = \lambda / (\pi w_0), \quad full\ angle \approx 2\theta \tag{4-3}$$

At fixed wavelength, halving the waist radius doubles the far-field half-angle. At the same time it reduces the Rayleigh range by a factor of four. For optical engineering this is a fundamental trade-off: a tighter focus provides a smaller spot but less longitudinal tolerance to axial displacement. Conversely, a beam that stays narrow for a long distance needs a larger waist or a shorter wavelength, subject to practical optics and laser-source limitations.

The trade-off can also be understood through logarithmic sensitivities. These derivatives are especially useful in a numerical study because they distinguish strong quadratic dependencies from merely linear ones. With the other parameters fixed, doubling the waist radius multiplies zR by four and divides θ by two. Doubling the wavelength halves zR and doubles θ.

$$\partial ln(zR) / \partial ln(w_0) = 2; \quad \partial ln(\theta) / \partial ln(w_0) = -1 \tag{4-4}$$

$$\partial ln(zR) / \partial ln(\lambda) = -1; \quad \partial ln(\theta) / \partial ln(\lambda) = 1 \tag{4-5}$$

The product $zR\theta = w_0$ follows from (3-7) and (4-3), linking the characteristic near-field distance and far-field spread directly to the waist size. The relation is an identity within the ideal paraxial model, not an independently measured constraint. Deviations in a real beam may be characterized with second-moment beam widths and the beam propagation ratio $M^2$ rather than by forcing a single perfect Gaussian fit [8].

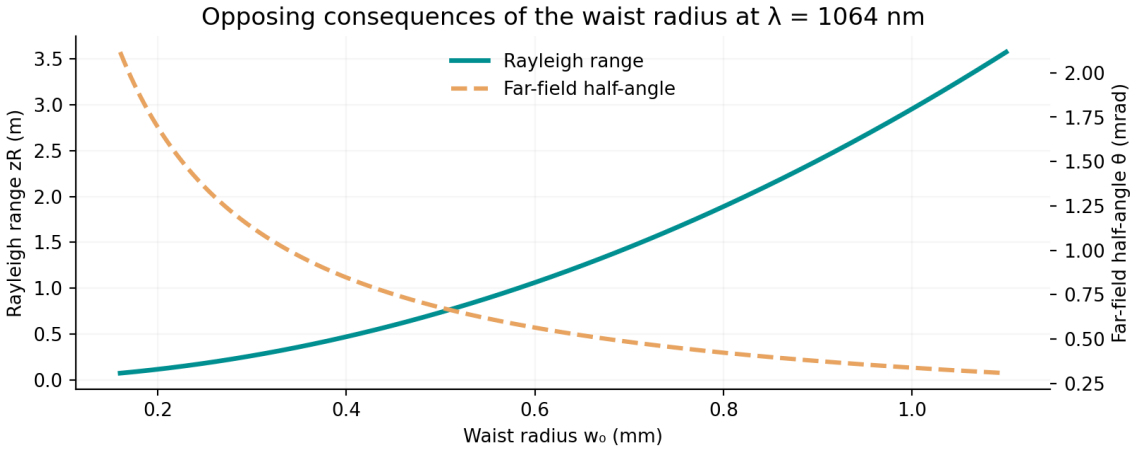



Figure 4.2 Calculated Rayleigh range and far-field divergence as ideal waist radius varies at fixed wavelength λ = 1064 nm. Values calculated from the ideal model.

## 4.3 Influence of wavelength, distance, and refractive medium

Wavelength enters the Rayleigh range inversely and the divergence directly when $w_0$ is fixed. For example, if the same 0.50-mm waist exists at 532 nm instead of 1064 nm in free space, its Rayleigh range doubles and its divergence half-angle halves. This comparison

isolates wavelength only; it does not imply that actual laser sources at these wavelengths necessarily produce identical spatial profiles or waist sizes.

Propagation distance is a separate design parameter. Near the waist, the spot stays within √2 of its minimum over $|z| \leq zR$; beyond that interval, radius growth progressively approaches the far-field linear limit. Because the transverse on-axis intensity scales as $w(z)^{-2}$ at constant power, both spot size and irradiance must be considered when evaluating a target plane. A relatively modest radius increase can correspond to a substantial reduction of peak intensity.

Within a uniform medium of refractive index n, the wavelength in the propagation expressions is $\lambda = \lambda_0/n$. The beam radius and propagation distance must be defined in that same medium. Passing through a refractive interface requires an appropriate optical transformation rather than simply reusing the free-space formula unchanged [1, 7].

## 4.4 Illustrative calculation and dimensional checks

As a numerical example, consider an ideal Gaussian beam with wavelength λ = 1064 nm, waist radius $w_0$ = 0.50 mm, and observation plane 1.00 m from the waist. These values are representative of a near-infrared laser arrangement and illustrate the unit conversions needed before comparing different beam parameters.

With SI units $\lambda = 1.064 \times 10^{-6}\, m$ and $w_0 = 5.0 \times 10^{-4}\, m$, Equation (3-7) gives $zR = \pi(5.0 \times 10^{-4}\, m)^2/(1.064 \times 10^{-6}\, m) \approx 0.7381\, m$. Equation (4-3) gives $\theta = 1.064 \times 10^{-6}\, m/[\pi(5.0 \times 10^{-4}\, m)] \approx 6.774 \times 10^{-4}\, rad$, or $0.6774\, mrad$. Then (3-6) predicts $w(1.00\, m) \approx 0.842\, mm$. Each result has the correct dimension: zR and w in meters and θ in radians.

Table 4.1 Verification case for ideal free-space Gaussian propagation.

| Quantity | Value | Interpretation |
|---|---|---|
| Input wavelength λ | 1064 nm | Near-infrared illustrative value |
| Waist radius $w_0$ | 0.50 mm | $1/e^2$ intensity radius at z = 0 |
| Rayleigh range zR | 0.738 m | $w = \sqrt{2}\, w_0$ at z = zR |
| Far-field half-angle θ | 0.677 mrad | Ideal asymptotic divergence |
| Radius at z = 1.00 m | 0.842 mm | Predicted $1/e^2$ radius |

The approximations in Equations (4-1) and (4-2) should not be applied indiscriminately at z around zR. At that distance neither $|z| \ll zR$ nor $|z| \gg zR$ is satisfied; the exact radius relation (3-6) should be used. Likewise, a diffraction-limited divergence result is meaningful only if the half-angle definition and ideal beam assumptions are maintained.

## 4.5 Chapter summary

This chapter quantified the near-field and far-field behaviors of the Gaussian envelope, derived divergence from the Rayleigh range, and identified the opposing parameter

dependence of beam confinement and spreading. The worked case provided a dimensional and numerical checkpoint for subsequent calculations. The next chapter introduces the complex parameter q and shows how lenses transform the beam radius and phase-front curvature together.

# Chapter 5: Beam Focusing and ABCD Matrix Theory

## 5.1 Why ray optics alone is insufficient

In paraxial ray optics, a propagation distance changes a ray's height according to its angle, while a thin lens changes the angle according to the ray's height. These operations can be expressed with simple transfer matrices. A Gaussian beam, however, has a finite transverse profile and a curved phase front. Its evolution therefore requires the matrix operation to act on a complex beam parameter, not merely on a single ray [2, 4].

For a ray vector (r, α), a first-order optical system in a fixed surrounding medium is represented by a real 2 × 2 matrix. In the usual paraxial convention, A, B, C, and D encode the linear dependence of output height and angle on the input quantities. Ideal lossless systems with the same refractive index at the input and output satisfy AD − BC = 1. This determinant relation is a useful algebraic check when combining components [2, 4].

$$[rout;\ \alpha out] \ = \ [A \ \ B;\ C \ \ D]\,[rin;\ \alpha in] \tag{5-1}$$

## 5.2 The complex Gaussian beam parameter

The central connection between ray transfer and Gaussian propagation is the complex beam parameter q, expressed in terms of the real curvature radius R and the Gaussian $1/e^2$ intensity radius w. One common convention is shown below. Both the real and imaginary parts of 1/q are physically significant, and the sign of the imaginary part depends on the chosen propagating-wave convention. The convention here is the one for which q = z + izR in free space when z = 0 is the waist [1, 2, 3, 4].

$$1/q(z) \ = \ 1/R(z) \ - \ i\,\lambda \,/\, [\pi\, w^2(z)] \tag{5-2}$$

$$q(z) \ = \ z \ + \ i\,zR \quad (distance\ z\ from\ an\ ideal\ waist) \tag{5-3}$$

Substitution of the free-space beam relations into (5-2) confirms (5-3). The real part tracks the propagation distance relative to the waist, while the imaginary part equals the Rayleigh range. These properties explain why a waist corresponds to a purely imaginary q parameter. The inverse representation also provides a practical recovery procedure: given q at a plane, compute 1/q and identify the real part with 1/R and the negative imaginary part with $\lambda/(\pi w^2)$.

Within the paraxial Gaussian approximation, q transforms under any first-order optical system by a bilinear fractional rule. Because optical systems are represented by the product of their component matrices, the same operation works for a single lens, a sequence of propagation spaces and lenses, or a more elaborate beam expander. The matrix order must

follow the order in which light encounters each component, with the first component matrix placed on the right in the total product [1, 2, 4].

$$qout = (A\,qin + B) / (C\,qin + D) \tag{5-4}$$

## 5.3 Free-space and thin-lens matrices

For propagation through free space over distance L, the ray height changes by L times the ray angle while the angle itself remains unchanged. For an ideal thin lens of focal length f, the ray height is continuous across the lens and its angle changes by $-r/f$. These two elementary systems have simple ABCD representations:

$$Mspace(L) = [1\ L;\ 0\ 1], \quad Mlens(f) = [1\ 0;\ -1/f\ 1] \tag{5-5}$$

Applying (5-4) to the free-space matrix gives $qout = qin + L$, as expected from Equation (5-3). Applying it to the thin-lens matrix gives a formula that changes the inverse curvature parameter by $-1/f$. Because a zero-thickness lens introduces phase without moving the transverse position of the ray, the Gaussian radius at the two immediately adjacent lens surfaces is the same in the ideal model. The wavefront curvature can nevertheless change abruptly.

$$1/qafter = 1/qbefore - 1/f \tag{5-6}$$

$$1/Rafter = 1/Rbefore - 1/f; \quad wafter = wbefore \tag{5-7}$$

This wave-optical interpretation prevents a common misconception: the beam does not physically contract to its new waist at the lens surface. Its phase front is altered there, and the smaller output waist forms only after subsequent propagation. A real lens also has finite thickness, finite aperture and aberrations; the thin-lens matrix is an approximation that neglects those effects [5].

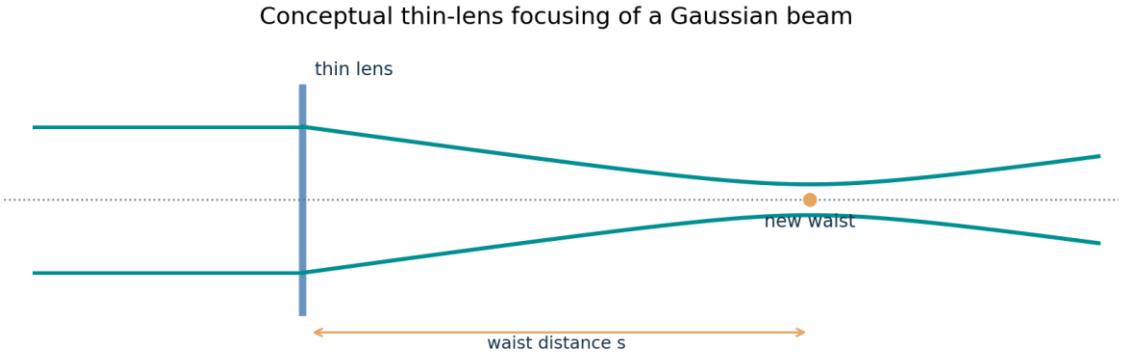



Figure 5.1 Schematic ideal-Gaussian transformation at a thin converging lens; waist position and envelope are illustrative, not drawn to scale.

## 5.4 Exact waist transformation through a single thin lens

Consider an input Gaussian beam with waist radius $w_0$ and Rayleigh range zR. Let its waist be a distance d upstream of a thin lens of positive focal length f. Then the incident q at the lens is $qin = d + izR$. Equation (5-6) gives the transformed q immediately after the lens; subsequent free-space propagation increases its real part by the traveled distance. The new waist occurs where the real part becomes zero. This construction yields exact first-order Gaussian formulas without assuming that the incident field is a plane wave [1, 5].

$$qin = d + i\,zR; \quad qafter = (d + i\,zR) / [1 - (d + i\,zR)/f] \tag{5-8}$$

Writing $qafter = -s + izR'$, where s is the distance from the lens to the output waist and zR′ is its Rayleigh range, gives closed-form expressions after separating real and imaginary parts. The sign of s must be interpreted physically: positive s denotes a real downstream waist, while negative s would denote a virtual waist upstream of the lens under this reference convention. The formulas are derived for matched input/output media and an ideal thin lens.

$$s = f\,[d(d - f) + zR^2] / [(d - f)^2 + zR^2] \tag{5-9}$$

$$zR' = f^2\,zR / [(d - f)^2 + zR^2] \tag{5-10}$$

$$w_0' = |f|\,w_0 / \sqrt{[(d - f)^2 + zR^2]} \tag{5-11}$$

To check these equations, set $d = f$, meaning that the input waist lies at the front focal plane of the lens. Then $s = f$ for all positive zR: the output waist lies at the back focal plane. The output waist size, however, is $f\,w_0/zR$ and is therefore not independent of the incident beam. This special result appears in classical Gaussian imaging theory and illustrates why geometric image position and Gaussian spot-size evolution should not be treated as the same variable [5].

Another useful case is d = 0: the incident waist coincides with the lens. Substitution gives $s = f\,zR^2/(f^2 + zR^2)$ and $w_0' = f\,w_0/\sqrt{(f^2 + zR^2)}$ for a positive lens. The new waist is then closer to the lens than its geometric focal length. For sufficiently large zR relative to f, the difference becomes small, but it is not exactly zero for a finite Gaussian beam.

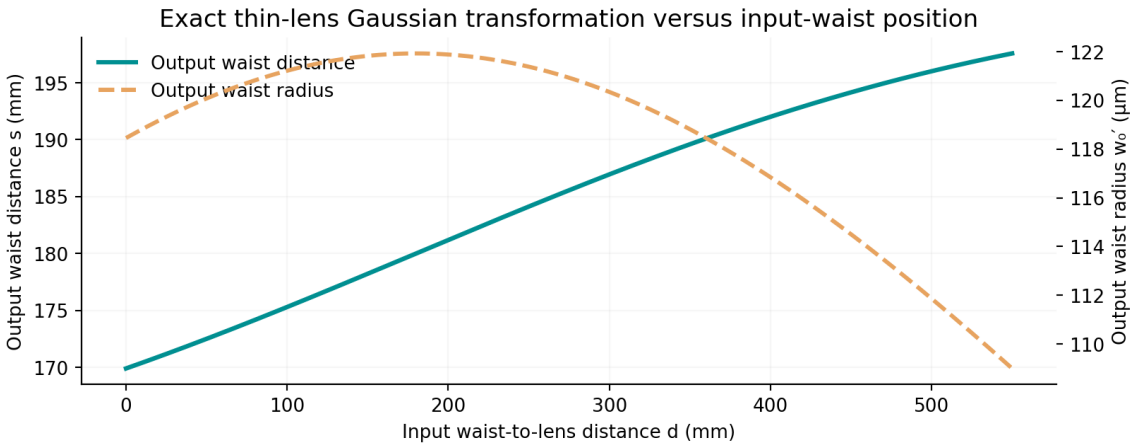



Figure 5.2 Ideal lens predictions from Equations (5-9) and (5-11) for λ = 1064 nm, input $w_0$ = 0.50 mm, f = 180 mm; varying d is a theoretical parameter study.

## 5.5 Approximate focusing formula and its conditions

Many engineering designs use an input beam that is nearly collimated at the lens, with local radius wL and very small phase-front curvature $1/RL$. When the corresponding Rayleigh range $zRL = \pi wL^2/\lambda$ is much larger than the focal length f, a strong focusing approximation follows. A positive lens then creates a new waist close to the back focal plane, with waist radius inversely proportional to the input radius [1, 5].

$$wfocus \approx \lambda f / (\pi\, wL) \quad if\ |f| \ll zRL\ and\ input\ curvature\ is\ weak \tag{5-12}$$

A representative example takes $\lambda = 1064\,nm$, $wL = 2.0\,mm$ and $f = 100\,mm$. The input Rayleigh range, for a planar phase front at that plane, is $zRL = \pi(0.002\,m)^2/(1.064 \times 10^{-6}\,m) \approx 11.81\,m$. Because $f/zRL \approx 0.00846$, the strong-focusing approximation is well justified. Equation (5-12) predicts a waist radius of approximately 16.9 μm. The exact formula gives an output waist location just short of 100 mm from the lens.

The formula does not apply equally well to a beam that is strongly converging or diverging at the lens, to an aperture-truncated beam, or to a focal spot approaching the diffraction limit of a strongly nonparaxial objective. A useful engineering report must therefore distinguish an approximate design estimate from the exact first-order Gaussian result and from experimentally attainable performance.

## 5.6 Cascaded optical components and beam collimation

For multiple ideal first-order elements, compute their ABCD matrices individually and multiply them in physical propagation order. A space of length L followed by a thin lens has a different total matrix from a lens followed by a space of the same length: matrix multiplication is not commutative. A beam expander often consists of two lenses separated by a chosen distance and produces a larger output beam radius together with a smaller far-

field divergence, provided the beam is not clipped and the optics are correctly positioned [1, 2].

$$Mtotal = M_{\mathrm{n}} M_{\mathrm{n-1}} \cdots M_2 M_1 \tag{5-13}$$

A simple ideal afocal telescope has focal lengths f1 and f2 separated by their algebraic sum $f1 + f2$, using an appropriate Galilean or Keplerian design. Its transverse magnification in magnitude is $|M| = |f2/f1|$. For nearly collimated, diffraction-limited Gaussian input and output beams, the output waist-scale radius grows approximately in proportion to $|M|$ and the far-field divergence shrinks by $|M|$. The beam-quality invariant is not defeated; the expansion trades beam width for reduced angular spread.

$$wout \approx |M| win; \quad \theta out \approx \theta in / |M| \quad (ideal\ afocal\ limit) \tag{5-14}$$

Collimation reduces the angular spread of a beam; it does not produce a perfectly parallel beam with finite radius. In the ideal Gaussian expression, zero divergence would require an infinitely large waist. A practical beam expander instead accepts a larger output diameter in exchange for a smaller far-field angle, within the limits of the optical apertures and alignment [1, 2].

## 5.7 Design constraints and interpretation

A small calculated focused radius is not a guarantee of a useful engineering focus. The actual intensity at the target also depends on laser power, lens transmission, aperture losses, alignment, surface contamination, aberrations, and beam quality. The paraxial calculation should be paired with clear input assumptions, for example stating whether wL is a measured 1/e² Gaussian radius or an effective width from a different convention. Where an ideal formula is used, the text should explicitly indicate that $M^2 = 1$ and that no aperturing or aberration is modeled [8].

The special value of the ABCD method is not the matrix notation itself but the ability to move consistently between a known incident beam, an optical element, and a predicted downstream waist. The method is both physically interpretable and numerically inexpensive, making it appropriate for the reproducible calculations that will form Chapter 6.

## 5.8 Chapter summary

This chapter introduced the complex q parameter, derived its transformation under first-order ABCD systems, and used the thin-lens relation to obtain explicit formulas for output waist location, Rayleigh range and waist radius. It clarified the limited conditions under which the simple focal-spot expression is accurate and explained collimation as a beam-width/divergence trade-off. Chapters 3–5 together provide the theoretical basis for the forthcoming numerical analysis and its practical interpretation.

# Chapter 6: Numerical Simulation and Image-Based Analysis

## 6.1 Two-study research design and data sources

Chapter 6 uses two different sources of evidence. Experiment 1 is a deterministic numerical simulation of an ideal circular $TEM_{00}$ beam. It uses the equations in Chapters 3–5 with specified wavelengths, initial beam waists, distances and thin-lens focal lengths; no optical measurements enter these calculations. Experiment 2 is an independent intensity-profile analysis of 14 existing JPEG images made available through the DataLab laser-beam-profiling tutorial [12]. In this second part, numerical results come from fitted image values rather than selected theoretical beam parameters.

Experiment 1 is divided into three computational studies: A changes wavelength at a fixed waist; B changes the input waist at a fixed wavelength; and C changes focal length and incident waist for an ideal thin lens placed at the incident waist plane. A further error calculation evaluates the simplified focusing formula. Experiment 2 uses the 14 ordered files TEM00_z_13.jpg through TEM00_z_80.jpg. The images were created outside this coursework; the code reads them as grayscale pixel arrays and records each profile fit. Table 6.1 describes the simulation input choices. Sections 6.7–6.10 describe the separate image-data workflow and its results.

Table 6.1 Deterministic parameter-study specification.

| Study | Varied inputs | Fixed quantities | Outputs |
|---|---|---|---|
| A: Wavelength | 532, 1064, 1550 nm | $w_0$ = 0.50 mm; z = 0–3 m | zR, θ, w(z) |
| B: Waist radius | 0.25, 0.50, 1.00 mm | λ = 1064 nm; z = 0–3 m | zR, θ, w(z) |
| C: Thin-lens focusing | f = 50–200 mm; $w_0$ = 0.5–2 mm | λ = 1064 nm; incident waist at lens | s, $w_0'$, approximation error |

Throughout the calculations the field is assumed to be monochromatic, circularly symmetric, diffraction-limited, and paraxial, with $M^2 = 1$. Propagation takes place in a uniform lossless medium approximated as free space. Lenses are modeled as ideal thin elements, without aperture clipping or aberration. These conditions keep the comparisons focused on the standard Gaussian relationships [1, 2, 5].

## 6.2 Simulation equations and numerical validation

The free-space model requires only the wavelength, waist radius and axial distance. The Rayleigh range is determined by Equation (3-7), the beam radius by Equation (3-6), and the far-field half-angle by Equation (4-3). The normalized quantities $w(z)/w_0$ and $z/zR$ allow comparison among beams without confusing differences in absolute size with differences in diffraction behavior. A numerical implementation can compute these functions using standard double-precision arithmetic; the model contains no fitted parameters and no statistical uncertainty.

$$zR = \pi w_0^2 / \lambda; \quad \theta = \lambda / (\pi w_0); \quad w(z) = w_0 \sqrt{[1 + (z/zR)^2]} \tag{6-1}$$

Several exact internal checks were applied before interpreting results. At $z = 0$, all evaluated curves satisfy $w(0) = w_0$. At $z = zR$, the computed radius equals $\sqrt{2}w_0$ to numerical precision. The far-field slope approaches $w_0/zR = \theta$. Moreover, doubling $w_0$ at unchanged wavelength must multiply zR by four while halving θ. These checkpoints detect mistaken unit conversions and incorrect use of full versus half divergence angle.

For the lens study, the selected geometry places the input beam waist in the lens plane, so the incident q parameter is purely imaginary: $qin = izR$. Applying the ideal thin-lens relation of Chapter 5 and propagating downstream gives the positive output-waist distance s, output minimum radius $w_0'$ and output Rayleigh range $zR'$. The exact formulas below avoid the geometric-optics assumption that the minimum radius always lies exactly one focal length behind the lens.

$$s = f\, zR^2 / (f^2 + zR^2); \quad w_0' = f\, w_0 / \sqrt{(f^2 + zR^2)} \tag{6-2}$$

$$zR' = f^2\, zR / (f^2 + zR^2); \quad wapprox = \lambda f / (\pi w_0) \tag{6-3}$$

For the stated incident-waist geometry, the relative excess of the approximate over the exact output waist is defined by $\varepsilon = 100 \times (wapprox - w_0')/w_0'$. Algebraic substitution produces $\varepsilon = 100[\sqrt{(1 + (f/zR)^2)} - 1]$. Thus $\varepsilon \geq 0$, and the approximation is only accurate when $f/zR$ is small. This expression isolates the approximation error from any additional, unmodeled limitations due to lens quality or beam nonideality.

$$\varepsilon\,(\%) = 100\,[\sqrt{(1 + (f / zR)^2)} - 1] \tag{6-4}$$

## 6.3 Study A: effect of wavelength on free-space propagation

With a common input waist $w_0 = 0.50\,mm$, shorter wavelengths produce a longer Rayleigh range and a smaller far-field divergence. Table 6.2 contains calculated quantities at three wavelengths representative of green, near-infrared and telecommunications-band optical beams. The wavelength values are used as mathematical comparison cases; no claims are made that real sources operating at these wavelengths automatically have equivalent input mode quality or power.

Table 6.2 Calculated ideal propagation for fixed $w_0$ = 0.50 mm.

| λ (nm) | zR (m) | θ (mrad) | w(1 m) (mm) | w(2 m) (mm) |
|---|---|---|---|---|
| 532 | 1.4763 | 0.3387 | 0.6039 | 0.8419 |
| 1064 | 0.7382 | 0.6774 | 0.8419 | 1.4441 |
| 1550 | 0.5067 | 0.9868 | 1.1062 | 2.0359 |

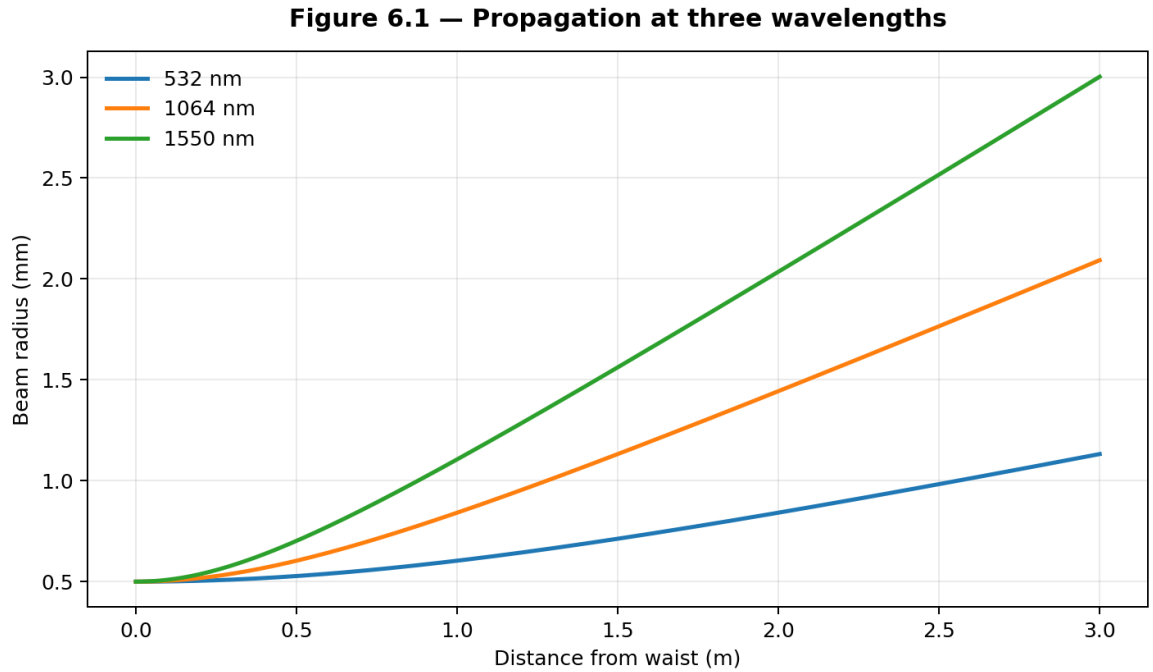



Figure 6.1 Computed radius-versus-distance trajectories for three wavelengths and a common input waist of 0.50 mm; theoretical predictions only.

At 1064 nm the numerical checkpoint is $zR \approx 0.738\,m$, $\theta \approx 0.677\,mrad$, and radius $w(1\,m) \approx 0.842\,mm$, consistent with the earlier calculation in Section 4.4. At 532 nm the Rayleigh range is doubled, while the ideal divergence is halved. At 1550 nm the opposite tendency appears. The spread is relatively subtle very close to the waist, but separation between the curves becomes larger as propagation distance increases. This is the wave-optical origin of an important engineering intuition: using a shorter wavelength improves ideal diffraction-limited directionality for a fixed launching waist.

A wavelength comparison must nevertheless hold other parameters fixed to remain meaningful. Changing a real source wavelength can also change optical coating performance, achievable waist size, detector response, scattering and atmospheric attenuation. The numerical curves answer a limited free-space diffraction question, not an all-inclusive ranking of wavelengths for a practical communication or processing task. That distinction becomes important in Chapter 7 [9, 10].

## 6.4 Study B: dependence on initial beam-waist radius

In the second sweep the wavelength is fixed at 1064 nm while three initial waist radii are compared. Although a smaller waist starts with a smaller spot, the accompanying larger divergence causes its radius to grow more rapidly. Thus a beam that is narrower at launch need not remain narrower at an arbitrarily distant observation plane. The crossover illustrates why waist radius and Rayleigh range must be selected together.

Table 6.3 Calculated waist-radius sweep for λ = 1064 nm.

| $w_0$ (mm) | zR (m) | θ (mrad) | w(1 m) (mm) | w(3 m) (mm) |
|---|---|---|---|---|
| 0.25 | 0.1845 | 1.3547 | 1.3776 | 4.0719 |
| 0.50 | 0.7382 | 0.6774 | 0.8419 | 2.0927 |
| 1.00 | 2.9526 | 0.3387 | 1.0558 | 1.4256 |

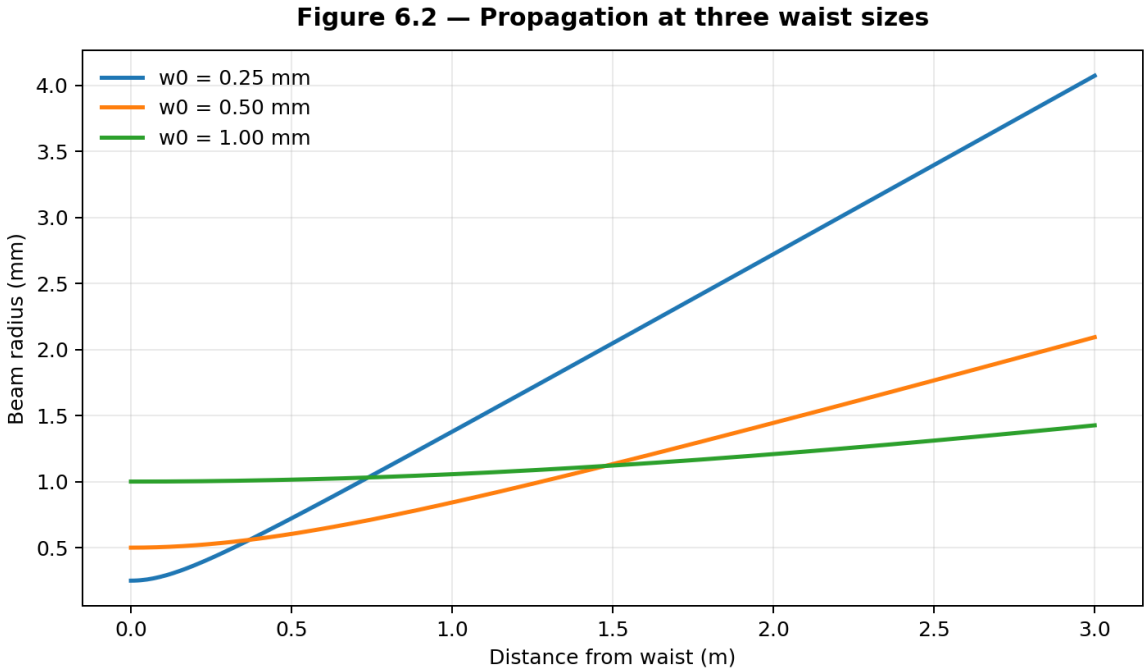



Figure 6.2 Absolute beam radius versus propagation distance for three initial waist radii at 1064 nm; smaller starting radius causes greater eventual spreading.

For example, increasing the initial waist from 0.25 mm to 0.50 mm multiplies the Rayleigh range by four and halves divergence. Increasing it again to 1.00 mm repeats the same relative change. Figure 6.3 shows that this benefit in confinement length is quadratic, not linear. Conversely, a radius of 1.00 mm at the waist is twice that of a 0.50-mm input, which may be undesirable where high irradiance or small delivery apertures are required. Therefore a wide beam should not be regarded as unconditionally superior; the correct design depends on the target distance and acceptable beam diameter.

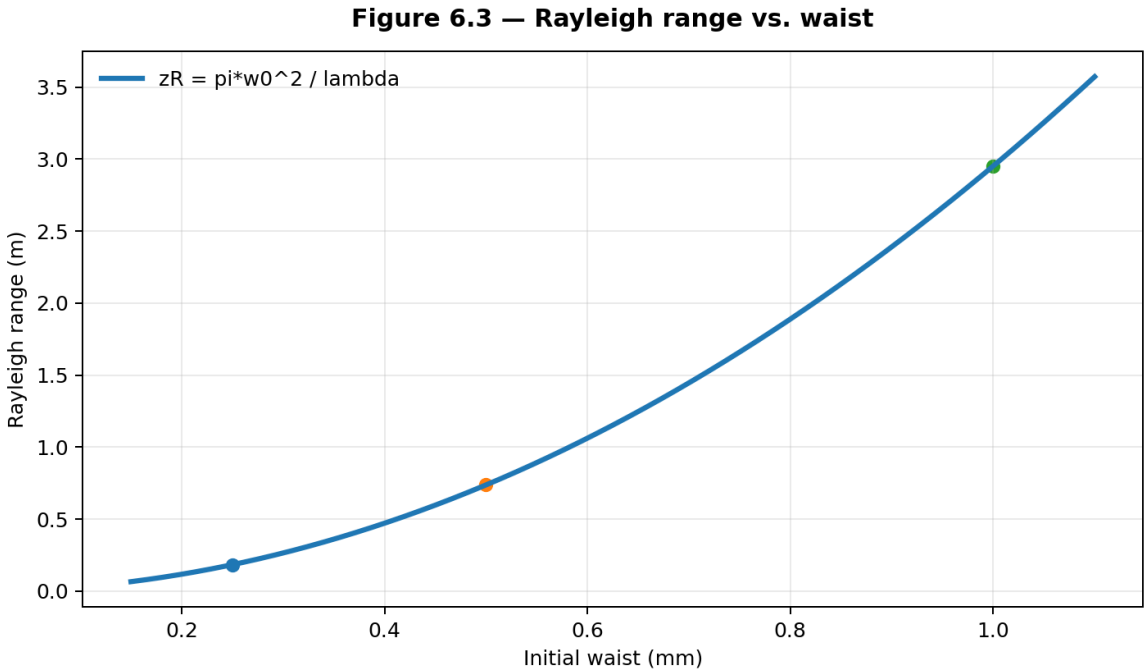



Figure 6.3 Ideal Rayleigh range versus initial waist radius at 1064 nm. The curve follows the quadratic expression $z_R = \pi w_0^2/\lambda$.

An alternative perspective is to consider the peak on-axis intensity at fixed optical power. At the waist $I(0,0) = 2P/(\pi w_0^2)$; doubling $w_0$ reduces this peak by a factor of four, even though it improves collimation. Smaller divergence and higher local intensity cannot generally be maximized simultaneously with a single ideal Gaussian beam at fixed power. This is a direct physical trade-off rather than an artifact of plotting conventions. It is central to the interpretation of focusing and beam-expansion devices [1].

## 6.5 Study C: lens focal length, input radius, and actual focus

The third sweep evaluates exact focusing using Equation (6-2) for the special but common case that the incident Gaussian beam is approximately flat-fronted at the lens because its waist is located there. In this case $z_R$ is calculated from the incident radius $w_0$; the output radius depends on both focal length and the incident Rayleigh range. A shorter focal length generally produces a smaller focus, whereas a larger incident beam decreases the minimum focused radius for the strong-focusing range. Both trends are conditional on the paraxial and finite-aperture assumptions.

Table 6.4 Exact theoretical focusing for input waist at lens, λ = 1064 nm.

| Input $w_0$ (mm) | f (mm) | s (mm) | Output $w_0'$ (μm) | Approx. error (%) |
|---|---|---|---|---|
| 0.5 | 50 | 49.77 | 33.79 | 0.229 |
| 0.5 | 100 | 98.20 | 67.12 | 0.913 |
| 0.5 | 200 | 186.32 | 130.76 | 3.606 |
| 1 | 50 | 49.99 | 16.93 | 0.014 |
| 1 | 100 | 99.89 | 33.85 | 0.057 |
| 1 | 200 | 199.09 | 67.58 | 0.229 |
| 2 | 50 | 50.00 | 8.47 | 0.001 |
| 2 | 100 | 99.99 | 16.93 | 0.004 |
| 2 | 200 | 199.94 | 33.86 | 0.014 |

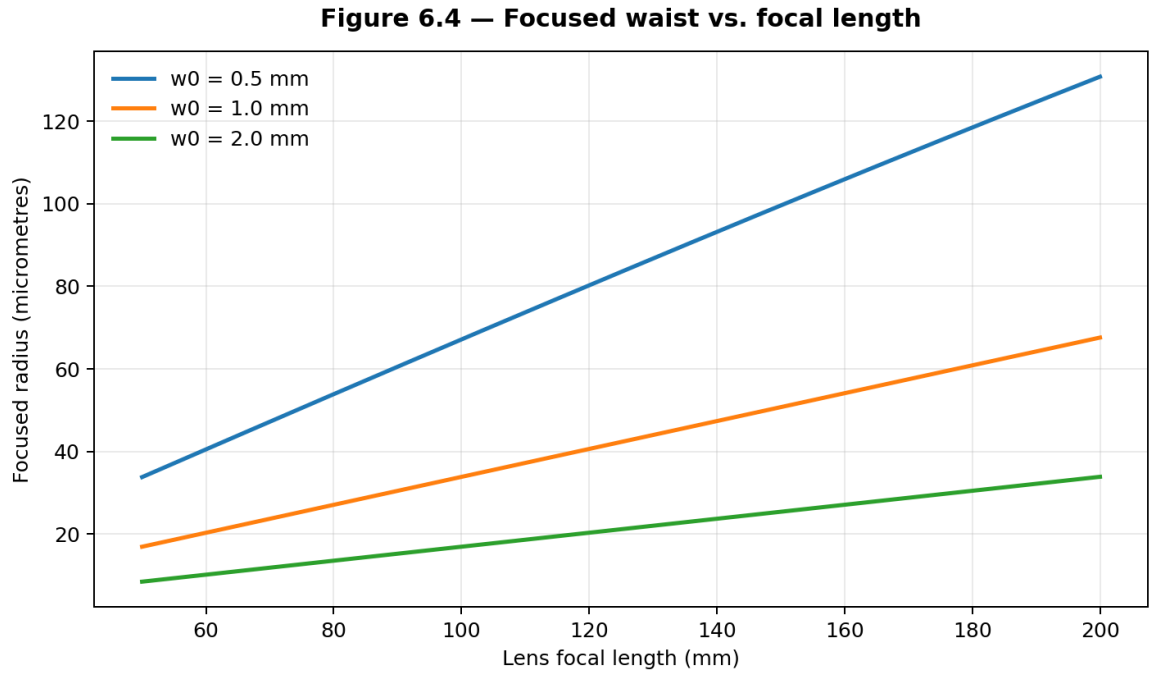



Figure 6.4 Predicted focused waist radius versus ideal thin-lens focal length for three incident waist radii. Input waists coincide with the lens plane.

For a 1064-nm beam with a 2.0-mm radius at the lens and a focal length of 100 mm, the predicted focused radius is approximately 16.93 μm. Its output waist forms about 99.99 mm beyond the lens. Here the incident Rayleigh range is approximately 11.81 m, so f is much smaller than zR and the simple focusing formula agrees closely with the exact result. For other focal lengths, the differences in waist position and radius can become significant.

A smaller predicted waist implies higher peak irradiance at fixed transmitted power. For instance, halving a focused radius while holding power unchanged raises the model peak intensity by a factor of four. However, an optical component must physically accommodate the incident beam. When the beam is enlarged to obtain a smaller focus and the lens diameter is not enlarged accordingly, truncation invalidates the ideal calculation. In

practice, desired spot size must therefore be evaluated with the clear-aperture radius, beam quality and allowable intensity at the lens.

## 6.6 Quantifying the error of the focal-spot approximation

Equation (6-4) quantifies the error introduced by $wapprox = \lambda f/(\pi w_0)$ when the incident waist is located at the lens. The error depends on the ratio $f/zR$. The estimate is reliable when that ratio is small; as the focal length approaches the Rayleigh range, the calculated spot radius becomes increasingly different from the exact Gaussian result. This condition should be checked before applying the approximation.

**Figure 6.5 — Error of approximate focusing formula**

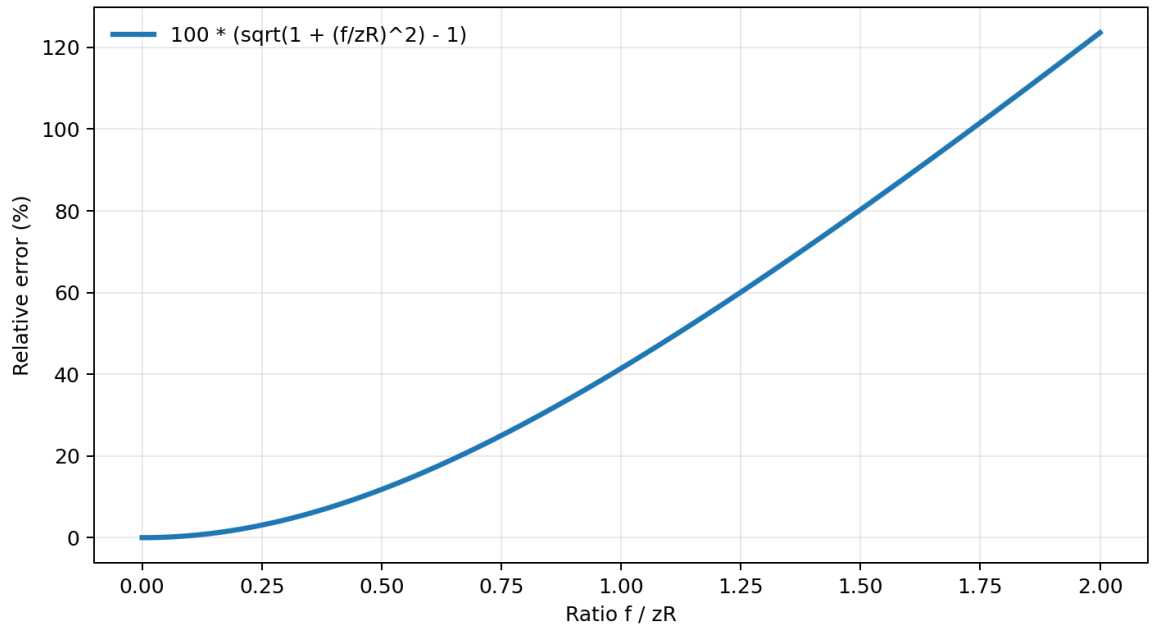



Figure 6.5 Analytical percentage overestimate of the approximate focused radius, expressed as a function of f/zR for the incident-waist-at-lens model.

Table 6.5 Focusing-approximation error as a function of f/zR.

| f/zR | Waist error (%) | s/f (exact) |
|---|---|---|
| 0.05 | 0.12 | 0.9975 |
| 0.10 | 0.50 | 0.9901 |
| 0.30 | 4.40 | 0.9174 |
| 0.50 | 11.80 | 0.8000 |
| 1.00 | 41.42 | 0.5000 |
| 2.00 | 123.61 | 0.2000 |

At $f/zR = 0.10$ the radius error is only about 0.50%. At $f/zR = 0.50$ the overestimate is about 11.80%, and at $f/zR = 1$ the estimate is larger than the exact radius by about 41.42%. The waist position is simultaneously displaced: $s/f = 1/[1 + (f/zR)^2]$. The simple rule that the waist lies at the back focal plane can be inaccurate even in ideal first-order Gaussian optics. This result provides the strongest applied interpretation

from the present numerical study: specify the incident wavefront/waist geometry before selecting or simplifying a lens-focusing formula [5].

## 6.7 Experiment 2: dataset provenance and purpose

The second study uses the DataLab Platform example titled “Laser beam profiling: measuring beam size along the propagation axis” [12]. DataLab supplies the TEM00_z_*.jpg image collection as tutorial data and demonstrates analysis of beam sizes from a sequence of images. I used the 14 files represented in the saved measurement table: TEM00_z_13.jpg, TEM00_z_18.jpg, TEM00_z_23.jpg, TEM00_z_30.jpg, and the files with labels 35, 40, 45, 50, 55, 60, 65, 70, 75 and 80. These are public example images, not images taken with my own camera or acquired in an original physical laser experiment.

This experiment asks a limited but useful question: do the bright transverse sections of those images have a shape that can be described well by a Gaussian intensity function, and how do the fitted apparent widths vary across the image sequence? The study does not assume that the filename number is a distance in millimeters or that the camera pixel scale is known. Although DataLab illustrates a separate calibration example in its tutorial, such a mapping is not independent acquisition metadata for this Python analysis. The labels are therefore plotted as ordering coordinates (arbitrary units), and the radii remain in pixels.

## 6.8 Image processing and Gaussian-fitting method

The analysis script analyze_beam_images.py opens each JPEG, converts it to an 8-bit grayscale intensity array, and estimates background level as the median pixel value sampled along the image border. The estimated background is subtracted and the remaining signal is clipped at zero to locate the beam. A preliminary centroid uses pixels with background-corrected signal at least 20% of the maximum. At that center, an average of 11 rows gives the horizontal profile, while an average of 11 columns gives the vertical profile. Averaging makes a single line less sensitive to small-scale noise, but it does not turn the sections into a full two-dimensional beam measurement.

Each one-dimensional gray-level profile is fitted independently to a Gaussian intensity curve with four free parameters: peak amplitude A, center $x_0$, $1/e^2$ intensity radius w, and constant offset B. The offset is included because the image intensity may have a residual camera background. SciPy nonlinear least squares (curve_fit) chooses the parameters by minimizing squared residuals. The initial width estimate comes from the half-maximum region; positive parameter bounds prevent nonphysical negative fitted radii. The fit uses the recorded gray levels, not calculated ideal intensity values.

Equation (6-5) One-dimensional Gaussian intensity fit used for the image sections.

| Profile model | Number |
|---|---|
| $I(x) = A\,exp\{-2[(x - x_0)/w]^2\} + B$ | (6-5) |

The radius w is the distance from the fitted center at which the fitted signal above its offset has decreased to $exp(-2)$ of its peak. For this definition, the full width at half maximum is $FWHM = w\sqrt{(2\,ln\,2)}$, rather than 2w. The reported fit statistic is $R^2 = 1 - SSE/SST$, where SSE is the sum of squared residuals and SST is the total squared deviation of the observed profile from its mean. An $R^2$ near one describes a close fit to that chosen one-dimensional model; it does not, by itself, establish Gaussian propagation along the axis or prove TEM‰ purity.

The output CSV records the filename, label, fitted w_x and w_y, derived FWHM values, fit $R^2$ for each section, the border-based background estimate, and the percentage of pixels with grayscale level at least 254. The last metric is a simple warning for near-saturation; it is not a detector calibration. Because JPEG encoding and 8-bit conversion can affect the profiles, the fitted radii should be viewed as image-based estimates. No physical pixel pitch, wavelength, lens configuration or independently known z coordinate is inferred from these data.

## 6.9 Results from the 14 laser-beam images

Table 6.6 reports the saved outputs for all 14 images. The horizontal fitted radius rises from 164.23 pixels in the first image to 297.96 pixels in the last; the vertical radius increases from 164.97 to 304.00 pixels. One small decrease occurs between the first and second horizontal estimates (164.23 to 163.83 pixels), so the correct interpretation is a strong overall broadening trend, not strict monotonic growth in both directions. The corresponding first-to-last ratios are 1.81 horizontally and 1.84 vertically.

Table 6.6 Gaussian fitting of DataLab image sections; radii in image pixels, labels in arbitrary units.

| Label | $w_x$ (px) | $w_y$ (px) | $R^2_x$ | $R^2_y$ | Near-sat. (%) |
|---|---|---|---|---|---|
| 13 | 164.23 | 164.97 | 0.9938 | 0.9957 | 0.0093 |
| 18 | 163.83 | 171.17 | 0.9937 | 0.9939 | 0.0000 |
| 23 | 172.98 | 178.57 | 0.9959 | 0.9955 | 0.0080 |
| 30 | 180.81 | 184.36 | 0.9945 | 0.9959 | 0.0031 |
| 35 | 189.70 | 191.60 | 0.9935 | 0.9954 | 0.0000 |
| 40 | 197.25 | 201.54 | 0.9951 | 0.9955 | 0.0021 |
| 45 | 209.55 | 211.93 | 0.9951 | 0.9956 | 0.0001 |
| 50 | 221.17 | 222.14 | 0.9946 | 0.9952 | 0.0004 |
| 55 | 232.67 | 238.22 | 0.9956 | 0.9943 | 0.0000 |
| 60 | 245.34 | 248.77 | 0.9952 | 0.9947 | 0.0003 |
| 65 | 257.93 | 264.04 | 0.9953 | 0.9939 | 0.0107 |
| 70 | 270.86 | 277.57 | 0.9950 | 0.9934 | 0.0000 |
| 75 | 284.45 | 293.22 | 0.9937 | 0.9927 | 0.0000 |
| 80 | 297.96 | 304.00 | 0.9953 | 0.9928 | 0.0000 |

Figure 6.6 uses the middle example TEM00_z_45.jpg to show an original grayscale image, the approximate beam center, and the corresponding horizontal and vertical profiles

with their fitted Gaussian curves. This visual check is useful because a numerical $R^2$ value alone cannot reveal all local residual structure. In this example the central lobe looks approximately symmetric, while the recorded background and small departures from the smooth fitting curve remain visible.

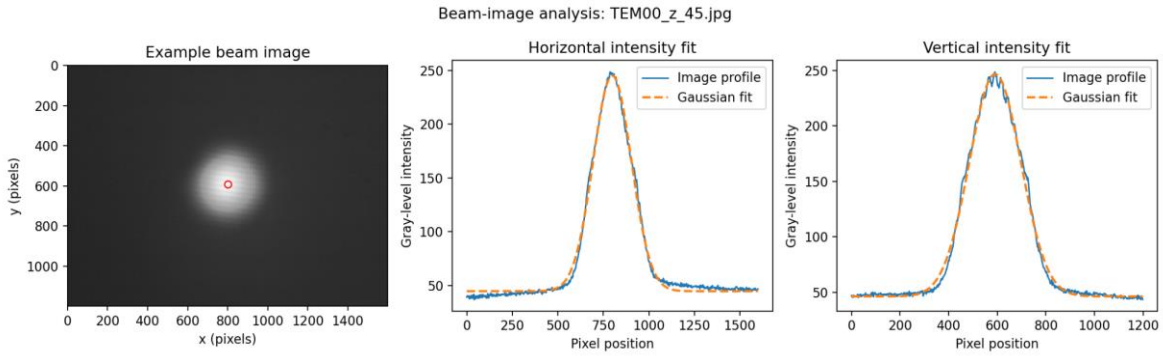



Figure 6.6 Grayscale example TEM00_z_45.jpg from the DataLab series with independently fitted horizontal and vertical Gaussian intensity sections; image positions and radii are in pixels.

Figure 6.7 summarizes the fitted widths against the numbers in the filenames. Both horizontal and vertical radii become larger toward the end of the series, and the vertical estimate is usually slightly larger. The mean relative difference $|w_x - w_y|/[(w_x + w_y)/2]$ across the 14 images is about 2.02%, with a largest observed difference of about 4.38%. These values suggest mild section-to-section asymmetry, but they cannot diagnose a particular cause such as astigmatism without the imaging geometry and acquisition calibration.

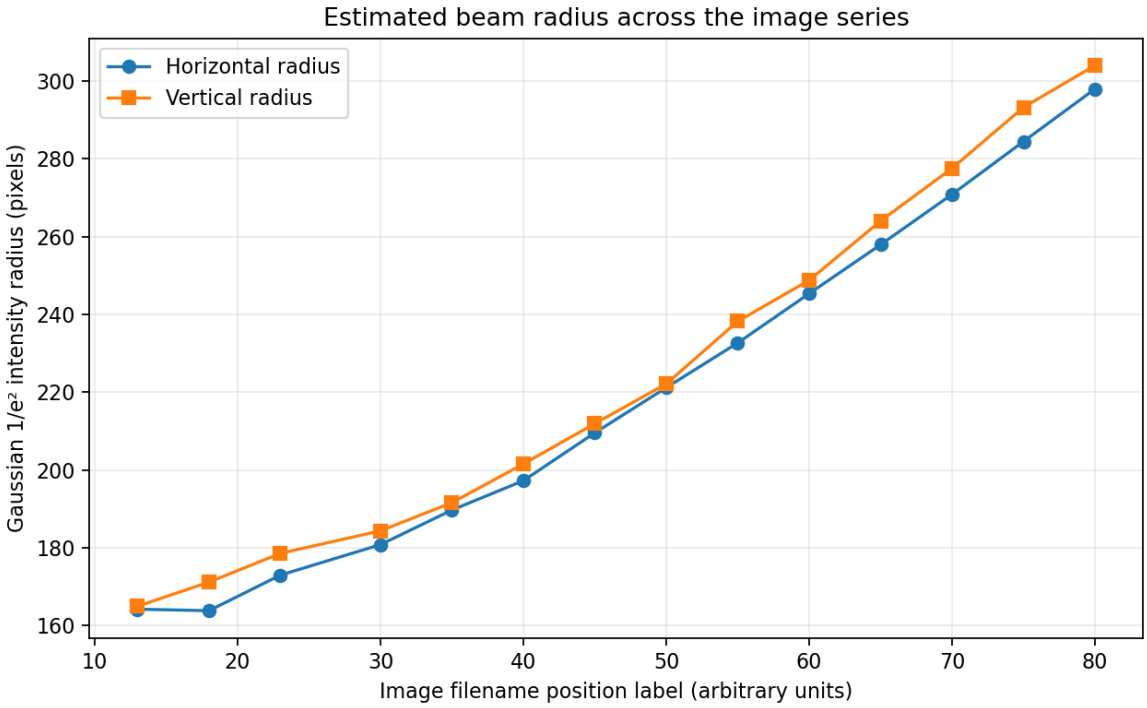



Figure 6.7 Horizontal and vertical fitted $1/e^2$ intensity radii across the DataLab image sequence; the filename label is an uncalibrated ordering coordinate.

The fitting quality is consistently high: the horizontal $R^2$ values range from 0.9935 to 0.9959, and the vertical values range from 0.9927 to 0.9959. The combined observed range is 0.9927–0.9959. Figure 6.8 shows that this result is consistent across the image sequence. High $R^2$ describes the effectiveness of this model for the sampled profiles; it does not quantify true instrument uncertainty or exclude systematic effects from offset selection and image compression.

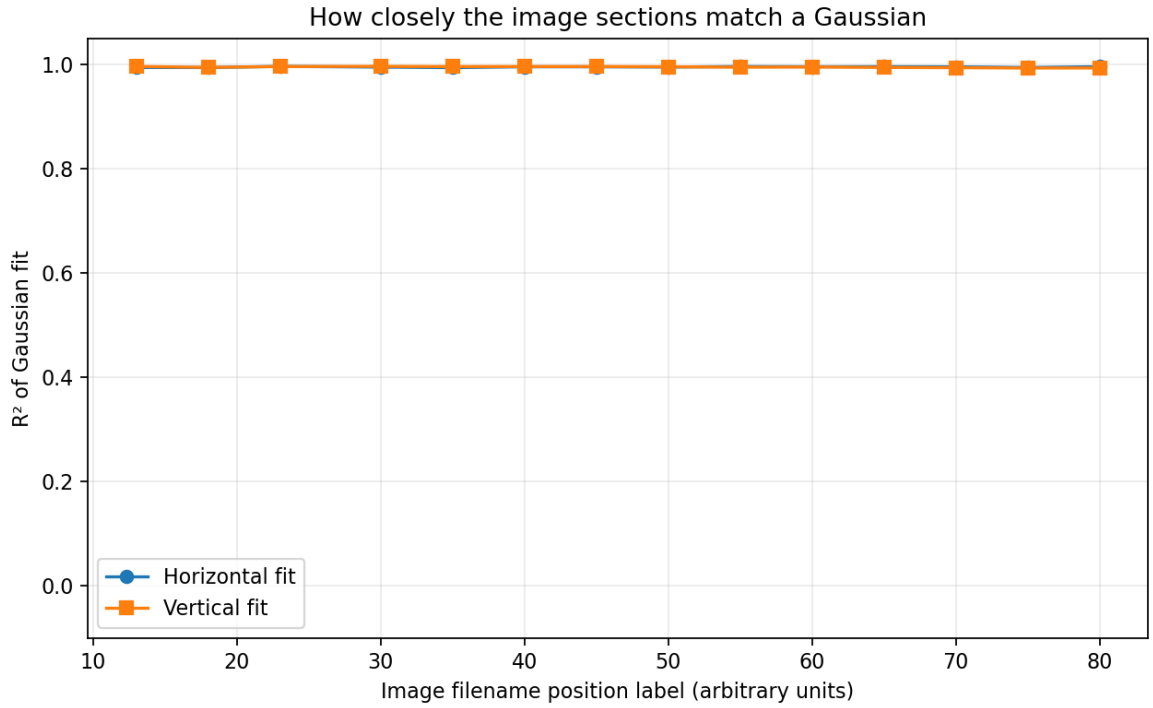



Figure 6.8 Coefficients of determination for independent horizontal and vertical Gaussian fits; these measure one-dimensional profile agreement, not laser beam-quality $M^2$.

Table 6.7 Descriptive statistics over the 14-image fitting results.

| Fitted quantity | Minimum | Mean | Maximum |
|---|---|---|---|
| Horizontal $1/e^2$ radius (px) | 163.83 | 220.62 | 297.96 |
| Vertical $1/e^2$ radius (px) | 164.97 | 225.15 | 304.00 |
| Horizontal profile $R^2$ | 0.9935 | 0.9947 | 0.9959 |
| Vertical profile $R^2$ | 0.9927 | 0.9946 | 0.9959 |
| Near-saturated pixels (%) | 0.0000 | 0.0024 | 0.0107 |

The percentage of pixels at or above grayscale value 254 is very low in the recorded files (maximum 0.0107%). This helps rule out widespread 8-bit saturation but cannot rule out clipping in an upstream camera exposure or tonal changes caused by JPEG processing. The border-estimated background varies from 15 to 44 gray levels across files, confirming why a flexible intensity offset was useful. Neither measurement alone establishes the precision of the camera or the original radiometric response.

## 6.10 Comparison between the theoretical and image studies

The theoretical and image results answer related but different questions. Experiment 1 gives model predictions in meters, millimeters, micrometers, and radians after the wavelength, initial waist and lens geometry are specified. Experiment 2 estimates the shape and apparent width of intensity profiles in pixels, starting from already recorded images. Both use the Gaussian function, but only the first uses a full propagation model with known physical parameters. In particular, a sequence of increasing fitted pixel widths does not

independently verify the Rayleigh range of the 1064-nm simulation or determine the position of the physical beam waist.

For a meaningful quantitative comparison, each image would need a calibrated transverse scale (mm per pixel), a known physical axial coordinate, and the relevant laser wavelength and optical configuration. Then a caustic fit could estimate propagation parameters from sufficiently distributed data around a waist; a standardized beam-width and $M^2$ determination would require additional measurement and processing conventions [8]. Without those inputs it is more accurate to describe Experiment 2 as Gaussian profile characterization than as validation of the full theoretical propagation law.

## 6.11 Discussion and boundaries of the two studies

Experiment 1 confirms the expected ideal-beam trends: smaller wavelengths reduce diffraction spreading for a fixed waist, and larger waists increase Rayleigh range while decreasing angular divergence. The focusing calculations also show why a focal-length-only approximation becomes inaccurate when the incident Rayleigh range is not sufficiently large.

Experiment 2 provides a different type of evidence. The DataLab images have profiles that are well described by Gaussian curves, with $R^2$ above 0.9927 for the fitted sections and a clear overall increase in apparent radius. This describes the recorded transverse patterns rather than establishing the physical propagation equation.

The simulation excludes real optical imperfections; the image study is limited by compression, finite gray-level resolution, background estimation, selected line sections and missing calibration. Its fitted pixel radii cannot establish millimeter-scale waist, Rayleigh range or $M^2$. The two studies can be compared quantitatively only after those physical measurements are available.

## 6.12 Chapter summary

Experiment 1 reproduced the expected Gaussian propagation and focusing dependencies for chosen ideal parameters. Experiment 2 fitted 14 existing beam images, obtaining high one-dimensional Gaussian $R^2$ values and a pronounced increase in pixel-scale radii across the sequence. Together they show why theoretical prediction and image-based measurement should be interpreted separately until physical spatial and axial calibrations are available.

# Chapter 7: Applications and Practical Limitations

## 7.1 Connecting Gaussian beam parameters to optical engineering

Gaussian propagation theory is useful because it turns the shape of a light beam into quantitative design parameters: a $1/e^2$ waist radius, a divergence half-angle, an axial Rayleigh range, a phase-front curvature, and the response to ideal lenses. In physical systems these quantities influence the radius of a power-delivery spot, allowable alignment error, collection efficiency, and the distance over which a beam remains adequately concentrated. The theory does not, by itself, determine useful performance; a system specification must also include optical power, losses, material interaction, wavefront quality and detector or target geometry [1, 7].

An important advantage of the Gaussian model is that its assumptions can be stated succinctly and tested. For a clean fundamental mode with a nearly circular profile, measuring the beam radius at several axial positions can support estimates of waist location and divergence. An optical engineer can then use the q law of Chapter 5 to choose elements for collimation or focusing. However, applying the same equations to a highly structured, truncated, multimode or strongly nonparaxial field without qualification can produce misleading predictions. The discussion below treats three representative settings as application contexts, not as directly tested case studies.

## 7.2 Precision laser processing and micromachining

Industrial material processing uses spatially concentrated laser energy for heating, melting, ablation, cutting and marking. A small focused spot can provide high spatial resolution and high irradiance, and Gaussian focusing offers a first design estimate for spot size and its axial sensitivity. For a circular ideal spot at a fixed transmitted power P, the peak irradiance is $Ipeak = 2P/(\pi w^2)$. If all other assumptions held, the 16.93-μm radius estimated in Section 6.5 would deliver substantially higher peak irradiance than a 33.86-μm radius at the same power. Such comparisons describe the incident beam, not the resulting kerf width or heat-affected zone.

The actual processed feature depends on pulse duration, scanning velocity, wavelength-dependent absorption, material thermal properties, threshold behavior and nonlinear mechanisms. Additional field effects from high numerical aperture, optical aberrations, spatter and thermal lensing may invalidate an ideal circular model. Recent work on laser additive-manufacturing processes notes that tailored spatial beam profiles can be advantageous relative to unmodified Gaussian illumination for some processing outcomes [11]. Therefore Gaussian is an analytical reference and common source mode, not a guarantee of the best possible spatial energy distribution for every material.

The most direct lesson for processing optics is to choose focused radius and depth of focus together. An extremely small minimum spot radius has a correspondingly short Rayleigh range, which can reduce axial positioning tolerance. A wider focus can decrease peak intensity but increase the depth over which approximately uniform spot size is maintained. The desired compromise depends on whether the priority is precision, throughput, processing stability or depth penetration. Numerical trends from Chapter 6 show the optical part of this compromise; a materials model is required for actual process optimization.

## 7.3 Free-space optical communications and beam delivery

In a free-space optical link, transmitter beam size and divergence contribute to geometric spreading and hence to the power distribution at a receiver. A larger outgoing beam radius can reduce far-field divergence if the beam is diffraction limited, but this benefit comes with greater transmitter-aperture requirements. The Gaussian envelope provides an ideal baseline estimate for the footprint at a nominal path distance. Receiver coupling, however, depends on collected power over the receiver aperture and on pointing accuracy, not only the radius on the optical axis.

Optical communication through an atmosphere encounters additional loss, turbulence, scintillation, beam wander and angular misalignment. Surveys of free-space optical communication explain why these propagation-medium and receiver effects must be modeled beyond simple diffraction [9, 10]. A wavelength offering lower ideal diffraction divergence may not produce a more reliable outdoor link once fog, absorption, atmospheric turbulence, eye-safety requirements and device constraints are included. Thus the 532–1550-nm numerical comparison is a statement about one controllable physics variable, not a practical transmission-efficiency forecast.

The baseline model is still valuable in early-stage design. An engineer can use Equations (3-6) and (4-3) to choose a candidate launch beam radius and estimate an ideal receiver-plane spot. Then a system-level model should add attenuation, aperture interception, tracking errors, wavefront distortions and detector sensitivity. The separation between the ideal propagation layer and the channel layer prevents unrealistic use of diffraction-limited predictions to claim field performance.

## 7.4 Imaging, metrology, and laboratory optical alignment

Laser-based measurement and imaging systems depend on controllable illumination geometry. In scanning instruments, a Gaussian focal spot may set a scale for lateral resolution and signal localization; in interferometric applications the complex wavefront curvature and Gouy phase can affect phase comparison and mode matching. Gaussian propagation formulas help specify a beam diameter at lenses, apertures and sample planes,

while ABCD matrices support first-order adjustment of optical relays or beam expanders [2, 6, 7].

A two-lens beam-expansion system illustrates the connection: increasing the beam diameter by a factor M reduces the ideal far-field angular spread by approximately 1/M under an afocal, diffraction-limited approximation. This does not violate diffraction physics because the reduced angle is achieved at the cost of a larger transverse extent. Lenses must also possess sufficient usable diameter; clipping the expanded beam can introduce diffraction artifacts and stray illumination.

For precision measurements, it is especially important to distinguish the theoretical $1/e^2$ Gaussian radius from a width measured using a camera or second-moment method. A measured beam may be elliptical, rotated, astigmatic or contaminated by background light. Its width must be defined consistently before comparison with the ideal calculations. The standardized approach in ISO 11146-1:2021 describes measurements of widths, divergence angles and propagation ratios for stigmatic and simple astigmatic beams [8].

## 7.5 Beam quality $M^2$ and the limits of the fundamental-mode formula

An ideal fundamental Gaussian has the minimum propagation product for the chosen width convention, often expressed as $M^2 = 1$. A real beam with $M^2$ greater than one generally exhibits larger divergence than an ideal fundamental Gaussian when compared at the same wavelength and waist width. The measured $M^2$ parameter cannot be inferred simply by looking at a bright central peak; meaningful determination requires defined beam-width measurements across axial positions and suitable fitting procedures [8]. The 14-image study does not measure $M^2$: its Gaussian fits describe single-plane horizontal and vertical sections, and its label series does not supply an independently calibrated optical caustic.

For a simple phenomenological model of an approximately stigmatic beam with a second-moment waist radius $w_0$ and propagation factor $M^2$, one often writes an effective Rayleigh range $zR,eff = \pi {w_0}^2/(M^2\lambda)$ and divergence $\theta eff = M^2\lambda/(\pi w_0)$. These are second-moment beam-propagation relations, not a claim that arbitrary multimode intensity remains exactly Gaussian. Higher-order modes may have non-Gaussian profiles, and aperture truncation can change both width and spatial power distribution.

$$zR,eff = \pi {w_0}^2 / (M^2 \lambda); \quad \theta eff = M^2 \lambda / (\pi w_0) \tag{7-1}$$

For illustrative scale only, an effective $M^2$ of 1.5 at the same 0.50-mm waist radius and 1064-nm wavelength would increase the divergence estimate from 0.677 mrad to approximately 1.016 mrad and reduce the effective Rayleigh range from 0.738 m to approximately 0.492 m. These numbers are not predictions for a particular laser and should not be interpreted as a complete non-Gaussian intensity simulation. They demonstrate why

a beam-quality measurement is needed before an ideal-lens design is transferred to real equipment.

## 7.6 Finite aperture, aberration, and focusing beyond the paraxial limit

The q-parameter rule assumes that an optical element is described by first-order paraxial transformations. A lens of finite clear radius a does not transmit the entire Gaussian profile: at a plane where the incident Gaussian radius is wL, the ideal encircled power fraction within a centered circular aperture is $1 - exp(-2a^2/wL^2)$. For a = wL only about 86.5% of input power lies inside the circular aperture, and the downstream field is no longer the untruncated Gaussian assumed by the simple q model. Larger clear apertures reduce such truncation, although real optical systems may impose size and cost limits.

Spherical aberration, coma, astigmatism, imperfect alignment and wavelength-dependent focal length can also enlarge or distort the focus. The approximation $wfocus \approx \lambda f/(\pi wL)$ is already conditional on a nearly collimated input. For a high-numerical-aperture objective producing a spot approaching the optical wavelength, scalar paraxial theory is not an adequate substitute for vectorial diffraction calculations. Such regimes are out of scope for the present dissertation and should not be inferred from Figures 6.4–6.5.

## 7.7 Engineering decision table

Table 7.1 Practical interpretation of the ideal Gaussian parameter trade-offs.

| Design goal | Ideal Gaussian strategy | Constraint not included in calculation |
|---|---|---|
| Long-distance delivery | Increase launch waist to reduce θ | Large optics, alignment, channel turbulence |
| Very small focal spot | Use larger input radius and/or shorter f | Aperture clipping, aberration, high NA |
| Longer near-waist region | Increase $w_0$ or decrease λ | Lower peak irradiance at fixed power |
| Accurate focusing estimate | Evaluate q and exact waist position | Input curvature and measured $M^2$ |
| Predictable optical measurement | Use defined width and waist conventions | Noise, asymmetry, detector calibration |

Table 7.1 emphasizes that a simple prescription such as “smaller spot is always better” is not an engineering design rule. The relevant quantity can be a spot radius, a divergence tolerance, energy delivered to a receiver, a thermal process threshold, or the stability of a measurement signal. Gaussian calculations support these decisions most effectively when linked to measurable quantities and the larger system requirements.

## 7.8 Camera-based beam profiling and interpretation

The image study adds a practical diagnostic perspective to the ideal propagation analysis. In beam profiling, the Gaussian model may be fitted to measured pixel intensity to obtain useful, reproducible descriptions of the beam center and transverse scale. This is

particularly helpful when a beam is nearly circular and the camera signal is not heavily saturated. The independent x and y estimates also make moderate shape asymmetry visible rather than hiding it in a single average radius.

However, the label “Gaussian” describes the fitted profile model, not a complete certification of laser quality. A high one-dimensional $R^2$ does not prove that the entire two-dimensional field contains only the $TEM_{00}$ mode or that its beam propagation factor is unity. Proper comparison with millimeter-based design tolerances requires calibrated camera sampling. Beam-width and $M^2$ reporting should follow an appropriate standardized measurement method such as ISO 11146-1, not simply a fitted line radius [8]. The distinction matters in laser alignment, optical communications, and tight focusing where an apparently clean profile may still be distorted by aberrations or pointing errors.

## 7.9 Chapter summary

Gaussian beams provide a foundational model for focusing, laser processing, free-space beam transmission and optical measurement. Its practical strengths are transparent parameter dependence and straightforward first-order lens modeling. Its main limitations are finite apertures, nonideal modes, aberrations, turbulence and high-NA/nonparaxial effects. The chapter established that real-world performance requires additional physical modeling and, where possible, standardized measurement rather than uncritical reliance on ideal equations.

# Chapter 8: Conclusion and Future Work

## 8.1 Summary of the dissertation

This dissertation examined Gaussian beam propagation and focusing by combining two studies. I first derived and evaluated the standard paraxial relations through three Python parameter sweeps. I then used Gaussian nonlinear fitting to examine 14 real image files from DataLab's published laser-beam example. The first study supplies physically specified theoretical predictions; the second supplies measured-image descriptions in pixels. Keeping those two outputs distinct is necessary for interpreting what has and has not been shown.

Numerical calculations were carried out for selected wavelengths, waist radii, and positive lens focal lengths. The resulting tables and curves show the changes in Rayleigh range, far-field divergence, beam radius, focused waist position, and focused spot size. Checks against limiting cases and unit consistency support the calculations and make their interpretation straightforward.

## 8.2 Main conclusions

The first conclusion is that ideal beam confinement, Rayleigh range and divergence are coupled. At fixed input waist, increasing wavelength increases far-field divergence and reduces Rayleigh range; at fixed wavelength, doubling the waist multiplies Rayleigh range by four while halving the divergence half-angle. The 1064-nm, 0.50-mm reference case gives $z_R = 0.738$ m and $\theta = 0.677$ mrad. These are predictions from the $TEM_{00}$ model, not measurements of a particular laser.

Second, beam focusing must account for the phase-front state of the incident field, not just the nominal focal length of a lens. The complex q parameter is a compact way to retain both radius and curvature. For the analyzed incident-waist-at-lens configuration, the exact output-waist distance and radius depend on the ratio of lens focal length to incident Rayleigh range. The popular approximate focus formula is excellent for a sufficiently large incident Rayleigh range but loses accuracy as $f/z_R$ increases.

Third, the numerical comparison offers a quantitative accuracy criterion: in the considered geometry the approximate formula overestimates the exact waist by about 0.50% at $f/z_R = 0.10$, 11.80% at 0.50, and 41.42% at 1.00. Therefore the incident waist location, local wavefront curvature and validity of the strong-focusing limit should be stated whenever a predicted focal spot is quoted. This is the main course-level methodological insight produced by the study, rather than a claim of new theoretical physics.

Fourth, ideal Gaussian formulas are useful design baselines rather than full engineering performance models. Laser material processing, free-space optical transmission and

precision metrology depend on additional factors—beam quality, aperture truncation, alignment, aberrations, absorption, turbulence, and detector response—that are excluded from the simple paraxial calculation. Careful labeling of ideal assumptions is essential for scientifically credible conclusions.

Fifth, the 14 published beam images show why profile fitting and propagation measurement should not be confused. The fitted intensity radii broaden from approximately 164 to 298 pixels horizontally and 165 to 304 pixels vertically, with fit $R^2$ values above 0.9927. Those results support Gaussian-like shape descriptions of the sampled image sections, but not a physical estimate of zR or $M^2$ because the required calibration data are absent.

## 8.3 Study limitations

The numerical study assumes a perfect circular Gaussian mode, free-space propagation and thin lenses without truncation or aberration. The image study uses published 8-bit JPEG photographs rather than an original calibrated optical experiment. Unknown pixel pitch, uncertain axial coordinate calibration, unspecified wavelength and acquisition conditions, background level variation, possible JPEG artifacts and one-dimensional section fitting prevent recovery of physical w(z), zR or $M^2$ from the current outputs. High $R^2$ values also do not prove mode purity or radiometric calibration.

The parameter sweeps are analytical calculations based on established Gaussian-beam theory. They indicate how the selected variables interact, but new measurements with verified distances, wavelengths, and camera scale would be needed to quantify performance in a particular system.

## 8.4 Future work

A meaningful next step would be to measure a beam at known axial positions with a calibrated profiler, record the camera pixel size, source wavelength and lens geometry, and repeat measurements near and on both sides of the waist. Using standardized second-moment widths where applicable would allow a beam caustic, Rayleigh range and $M^2$ to be estimated with stated uncertainties [8]. The images in Experiment 2 could also be reprocessed using full two-dimensional fitting and residual maps to examine sensitivity to background and line-section selection.

A second extension would implement Fourier-optics propagation for clipped or apodized beams. Such a model can explicitly predict diffraction-induced sidelobes beyond a finite aperture, extending the current geometric q description into regimes where the transmitted beam is no longer an exact Gaussian. For very tight focusing, a vectorial high-numerical-aperture formulation would be appropriate. For longer outdoor paths, a

turbulence/pointing model could be coupled to the free-space baseline to study optical-link reliability [9, 10].

A third extension could optimize a compound lens system for conflicting objectives such as minimum spot size, specified working distance, allowable aperture and depth of focus. The system may be described at first by cascaded ABCD matrices, with later refinements incorporating aberrations and measured beam quality. Such extensions would transform the present reproducible teaching-oriented analysis into a more comprehensive optical engineering design exercise.

## 8.5 Final conclusion

Gaussian-beam theory is useful both for predicting beam behavior from optical parameters and for describing transverse intensity measurements. The numerical study demonstrates the diffraction trade-offs and the limitations of a simplified focus formula; the existing-image study demonstrates that Gaussian functions closely match selected recorded profiles and that their apparent widths broaden over the sequence. The strongest conclusion is not that the two experiments are numerically identical, but that their results become physically comparable only when the image measurements are calibrated and their assumptions are made explicit.

# References

[1] SIEGMAN A E. Lasers[M]. Mill Valley, CA: University Science Books, 1986.

[2] KOGELNIK H, LI T. Laser beams and resonators[J]. Applied Optics, 1966, 5(10): 1550-1567. DOI: 10.1364/AO.5.001550.

[3] KOGELNIK H. On the propagation of Gaussian beams of light through lenslike media including those with a loss or gain variation[J]. Applied Optics, 1965, 4(12): 1562-1569. DOI: 10.1364/AO.4.001562.

[4] KOGELNIK H. Imaging of optical modes—resonators with internal lenses[J]. Bell System Technical Journal, 1965, 44(3): 455-494. DOI: 10.1002/j.1538-7305.1965.tb01672.x.

[5] SELF S A. Focusing of spherical Gaussian beams[J]. Applied Optics, 1983, 22(5): 658-661. DOI: 10.1364/AO.22.000658.

[6] GOODMAN J W. Introduction to Fourier Optics[M]. 4th ed. New York: W. H. Freeman, 2017.

[7] SALEH B E A, TEICH M C. Fundamentals of Photonics[M]. 3rd ed. Hoboken, NJ: Wiley, 2019.

[8] ISO 11146-1:2021. Lasers and laser-related equipment—Test methods for laser beam widths, divergence angles and beam propagation ratios—Part 1: Stigmatic and simple astigmatic beams[S]. Geneva: International Organization for Standardization, 2021.

[9] KHALIGHI M A, UYSAL M. Survey on free space optical communication: A communication theory perspective[J]. IEEE Communications Surveys & Tutorials, 2014, 16(4): 2231-2258. DOI: 10.1109/COMST.2014.2329501.

[10] PHUCHORTHAM S, SABIT H. A review of the structure of free-space optical channel models: Physical meaning, assumptions, and atmospheric conditions[J]. Photonics, 2026, 13(5): 425. DOI: 10.3390/photonics13050425.

[11] LIAN R, XU X, ZHAO K, et al. Advancements in laser processes for additive manufacturing: Techniques and future directions[J]. Journal of Materials Research and Technology, 2026, 41: 2428-2451. DOI: 10.1016/j.jmrt.2026.01.173.

[12] DATALAB PLATFORM. Laser beam profiling: measuring beam size along the propagation axis [EB/OL]. DataLab documentation. https://datalab-platform.com/en/intro/tutorials/laser_beam.html (accessed 3 November 2026).

# Appendix A: Reproducibility Notes

## A.1 Analytical calculation sequence

This appendix records sufficient detail to regenerate the Chapter 6 results. All length inputs should first be converted to meters: $nanometers \times 10^{-9}$, $millimeters \times 10^{-3}$, $micrometers \times 10^{-6}$. For a chosen wavelength λ and waist radius $w_0$, evaluate the Rayleigh range $zR = \pi {w_0}^2/\lambda$ and divergence $\theta = \lambda/(\pi w_0)$. At a prescribed distance z, calculate $w(z) = w_0\sqrt{[1 + (z/zR)^2]}$. Results in meters or radians may then be scaled to the tabulated display units.

For an input waist placed at the lens plane, calculate zR and then the exact output values $s = fzR^2/(f^2 + zR^2)$, $w_0' = fw_0/\sqrt{(f^2 + zR^2)}$, and $zR' = f^2zR/(f^2 + zR^2)$. Evaluate the approximation $wapprox = \lambda f/(\pi w_0)$, and calculate its relative radius error as $100(wapprox/w_0' - 1)$. Figures use smooth evenly spaced parameter arrays; these are interpolated model evaluations, not independent measurements.

## A.2 Minimal reproducibility pseudocode

```
def gaussian(lambda_m, w0_m, z_m):
  zR = pi * w0_m ** 2 / lambda_m
  theta = lambda_m / (pi * w0_m)
  w = w0_m * sqrt(1 + (z_m / zR) ** 2)
  return zR, theta, w

def focus(lambda_m, w0_m, f_m):
  zR = pi * w0_m ** 2 / lambda_m
  s = f_m * zR ** 2 / (f_m ** 2 + zR ** 2)
  wnew = f_m * w0_m / sqrt(f_m ** 2 + zR ** 2)
  wapprox = lambda_m * f_m / (pi * w0_m)
  return s, wnew, 100 * (wapprox / wnew - 1)
```

The expressions above describe an ideal diffraction-limited Gaussian beam and a thin lens. They reproduce the values used in Chapter 6; comparisons with measured laser systems would require additional information on beam quality, aberration, and uncertainty.

# Appendix B: Image Analysis Reproducibility

## B.1 Input and output files

Dataset: DataLab Platform tutorial "Laser beam profiling: measuring beam size along the propagation axis" [12]. Source images: 14 TEM00_z_*.jpg files, ordered by numeric filename label. The external images are third-party tutorial examples; they are not original photographs collected for this coursework. The script can read the published laser_beam.zip archive or extracted image files. No image pixels are artificially generated for the fit. The saved results table contains 14 processed records.

The Python program is analyze_beam_images.py (NumPy, SciPy, Pillow, Matplotlib). The principal output is beam_measurements.csv, with fields image_order, image_name, image_label, w_x_px, w_y_px, fwhm_x_px, fwhm_y_px, r2_x, r2_y, background_graylevel and saturated_pixels_pct. The three figures are example_image_and_fits.png, beam_width_across_images.png and fit_quality_across_images.png. Experiment 1 uses experiment.py and its separate numerical plots and CSV tables.

## B.2 Processing sequence and units

For each image: (1) open the JPEG and convert it to grayscale; (2) calculate the median sampled image-border gray level; (3) subtract background and find a preliminary intensity-weighted center above a 20%-of-peak threshold; (4) average 11 rows or 11 columns around the estimated center; (5) fit each section to A $exp[-2((x-x_0)/w)^2] + B$ with constrained nonlinear least squares; (6) report fitted w, $FWHM = w\sqrt{(2\,ln2)}$, coefficient of determination $R^2$ and near-saturated-pixel percentage; (7) plot both radii and the $R^2$ values against image label.

The image_label field preserves the number in the JPEG filename and is shown in arbitrary units. The fitted radii are expressed in pixels, and the graphs must not be relabeled in millimeters without spatial calibration. Physical laser wavelength, camera pixel pitch, and independently known axial distances are not inferred from the image labels. A Gaussian fit is not a measurement of $M^2$; no calibrated physical Rayleigh range is reported for these files.

## B.3 Verification and reproducibility limitations

The numerical CSV tables for Experiment 1 can be reproduced directly from the stated Gaussian equations and input parameters. The CSV and plots for Experiment 2 reflect the 14 analyzed JPEGs available in the saved prior output. A full new end-to-end rerun of the JPEG fitting requires downloading the original DataLab image archive and using the

analysis script with the same libraries and preprocessing settings. The original photographs are not embedded in this dissertation, except for the illustrative view reproduced inside Figure 6.6. When comparing results from separate program runs, small differences can arise from library versions, image decoding or fitting tolerances.